

Open-query games on ordinal spaces

A negative solution to the topological-sum problem,
with an exact Cantor–Bendixson classification

Research report

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For the convergent ordinal sequence $S = [0, \omega]$,

$$\text{sm}(S) = 1, \quad \text{sm}(S \sqcup S) = 2.$$

More generally, for every integer $k \geq 1$, the compact ordinal space

$$K_k = [0, \omega^{2^k-1}]$$

satisfies $\text{sm}(K_k) = k$ and $\text{sm}(K_k \sqcup K_k) = k + 1$.

Complete ordinary mathematical proofs are included. The archive also contains an independent finite-poset verifier, exact data, and a machine-readable strategy certificate. The infinite-space results do not depend on numerical experiments. No claim of independent peer review or formal proof-assistant verification is made.

Part II, added on 29 September 2026 (batch 52), extends the theory from a membership bit to q output labels: sharp synchronization with overhead $\lceil \log_2 q \rceil$, the exact q -label invariant of convergent towers, and a polynomial/NP-complete dichotomy between two and three labels.

Abstract

We answer negatively Problem 1.3 of Chiozini, Csernák, and Soukup’s *Gamification of the T_0 -pseudoweight via cut-and-choose games on topological spaces*, in the version available as arXiv:2510.05754v3. The counterexample consists of two convergent sequences, so it lies entirely within compact metrizable ordinal spaces.

The main further result is an exact formula for every nonempty Hausdorff space of finite Cantor–Bendixson height. If $X^{(h)} \neq \emptyset$ and $X^{(h+1)} = \emptyset$, then

$$\text{sm}(X) = \begin{cases} \lceil \log_2(h+1) \rceil, & |X^{(h)}| = 1, \\ \lceil \log_2(h+2) \rceil, & |X^{(h)}| \geq 2. \end{cases}$$

The proof identifies finite open-query complexity with the logarithm of the minimum length of an alternating closed-layer decomposition. A canonical alternating-closure procedure computes that length and retains the orientation information lost by the scalar invariant sm .

Consequences include sharp one-move failures of the proposed sum identity at every positive finite level, exact finite-product formulae, a classification on countable scattered Hausdorff spaces, and a quadratic-time target algorithm for finite posets. Independent minimax computations verify the finite normal form on all naturally labelled posets through six elements and all their targets: 320,866 tests in total.

Addition of 29 September 2026 (batch 52). Part II, from a later manuscript, lets the Seeker determine one of $q \geq 2$ labels instead of a membership bit. Optimal depth is the ceiling of the binary logarithm of a minimum number of ordered open-suffix layers; on posets this number is a shortest common supersequence of chain words. Components with at most m layers need at most $(q-1)m+1$ jointly, so combining them can cost exactly $\lceil \log_2 q \rceil$ extra questions, sharply, already for fixed colorings of compact countable metrizable spaces. A space of last derivative rank h with s colors on its last derivative needs at most $(q-1)h+s$ layers, sharply; the worst case over q -colorings of t convergent towers of rank $m-1$ is $\lceil \log_2((q-1)(m-1) + \min(q, t)) \rceil$, which at $q=2$ agrees with the formula above for $h=m-1$. Exact optimization is polynomial for two labels and NP-complete for three, with a polynomial algorithm within $\lceil \log_2(q-1) \rceil$ questions of optimal.

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1 The selected problem and the status of the answer

In the set-membership game, questions are restricted to open subsets of a space, and the aim is to decide membership in a fixed target subset. The invariant $\text{sm}(X)$ is the worst, over all targets, of the least ordinal number of questions needed. Chiozini, Csernák, and Soukup ask whether

$$\text{sm}\left(\bigsqcup_{i \in I} X_i\right) \stackrel{?}{=} \sup_{i \in I} \text{sm}(X_i). \quad (1.1)$$

This is their Problem 1.3; their Theorem 3.4 gives the upper bound consisting of the right-hand side followed by one additional move [1].

The answer to (1.1) is *no*. The proof in the next section is short and does not depend on any later theorem. In particular, the counterexample is not an artifact of non-Hausdorff topology, uncountable spaces, or an ambiguity about transfinite play. It uses a two-move game on a countable compact metrizable space. Repeating the same component countably many times also gives a counterexample when the index set is required to be infinite.

The more substantial development is the classification in theorem 6.2. Its mechanism explains exactly why the counterexample works. Different components may require opposite interpretations of the same open-set answer. A scalar question count forgets that orientation, while a pair of alternating-closure lengths preserves it.

Status and provenance. The arXiv record retrieved for this report lists version 3, dated 29 May 2026; the question remains explicitly printed there [1]. The report gives a negative solution to that stated question, not a solution to every open problem in the paper. A targeted literature search found no intervening resolution of this question. This is a search statement, not a proof of priority. The closed-layer machinery belongs to the classical setting of Hausdorff difference hierarchies; we do not claim that framework as new [2]. All lemmas needed for the results here are proved below.

[Added 29 September 2026, batch 52] **Part II.** Sections 1–13 and Appendices A–B are this report as of 19 September 2026 (Part I). Part II (section 14) is a later manuscript that extends the game from a membership bit to q output labels and reproduces Part I’s values at $q = 2$ (section 14.4). It does not change any statement of Part I.

2 A compact ordinal counterexample

Let $S = [0, \omega]$ have its order topology. Thus every natural number is isolated, and the neighborhoods of ω contain a final segment of $\mathbb{N}$. We use zero questions for a constant target: a subset equal to the whole space or to the empty set.

Lemma 2.1 (The one-question criterion). *For a subset A of any topological space X , membership in A can be decided with at most one open-set question if and only if A is open or closed.*

Proof. An open target is itself a legal question. A closed target can be tested by asking its open complement. Conversely, suppose the first question is U . The only two possible final states are U and $X \setminus U$. Both must be monochromatic for membership in A . If they have the same color, A is constant. Otherwise $A = U$ or $A = X \setminus U$. $\square$

Proposition 2.2. *For $S = [0, \omega]$,*

$$\text{sm}(S) = 1, \quad \text{sm}(S \sqcup S) = 2. \quad (2.1)$$

Proof. Every subset of S missing ω is open, since it consists entirely of isolated points. Every subset containing ω is closed, since its complement consists of isolated points. Hence every target takes at most one question. There are nonconstant targets, so $\text{sm}(S) = 1$.

Write S_0, S_1 for two disjoint copies, with limit points p_0, p_1 , and put

$$A = \{p_0\} \sqcup (S_1 \setminus \{p_1\}).$$

This set is not open: it contains p_0 but no isolated point of S_0 . It is not closed: it contains all isolated points of S_1 , but omits their limit p_1 . By lemma 2.1, one question cannot suffice.

Two questions suffice for *every* target B . First ask the clopen set S_0 , thereby determining which copy contains the remaining candidates. Inside that copy, B is open or closed, so one more open question decides membership. An open set in a component is open in the disjoint union. $\square$

The fact that spaces with at most one accumulation point have one-question membership complexity is also noted in [1, Section 7]; the failure of (1.1) comes from putting two such components together.

Corollary 2.3. *The proposed identity fails even when every component and the whole sum are compact, countable, metrizable, zero-dimensional, and homeomorphic to ordinal spaces. Specifically,*

$$S \sqcup S \cong [0, \omega \cdot 2], \quad \text{sm}([0, \omega \cdot 2]) = 2.$$

Proof. The two clopen intervals $[0, \omega]$ and $(\omega, \omega \cdot 2]$ partition $[0, \omega \cdot 2]$. Each has order type $\omega + 1$. The stated topological properties can also be seen directly from two disjoint convergent sequences. $\square$

Proposition 2.4 (An infinite index set). *For every cardinal $\kappa \geq 2$,*

$$\text{sm}\left(\bigsqcup_{i < \kappa} S\right) = 2.$$

Proof. The lower bound follows by restricting to two components. For the upper bound, fix a target A , and split the component indices into two classes so that A_i is open on the first class and closed on the second; when both apply, choose either class. One question asks the union of the components in the first class. A second question asks the open set whose trace is A_i on the first class and $S_i \setminus A_i$ on the second. The two answers determine the color. $\square$

3 The game, subspaces, and ordinal conventions

All ordinal intervals in this report carry their order topology. The notation $[0, \gamma]$ denotes a space whose underlying ordered set has order type $\gamma + 1$. A topological sum $X \sqcup Y$ is not being confused with an ordinal sum; when they are homeomorphic, this is stated separately.

Definition 3.1 (Open-query membership game). Fix $A \subseteq X$ and an ordinal β . At stage $\xi < \beta$, the Seeker chooses an open set U_ξ . The Hider answers $e_\xi \in \{0, 1\}$. Set

$$W_\eta = \bigcap_{\xi < \eta} \begin{cases} U_\xi, & e_\xi = 1, \\ X \setminus U_\xi, & e_\xi = 0. \end{cases}$$

The Seeker wins if $W_\beta \subseteq A$ or $W_\beta \cap A = \emptyset$. Write $\text{sm}(A, X)$ for the least winning length and $\text{sm}(X) = \sup_{A \subseteq X} \text{sm}(A, X)$, as in [1].

For ordinal-valued invariants we work with T_0 spaces, where a winning length exists: asking all open sets separates points. A statement of the form $\text{sm}(A, X) \leq d$ can also be read on an arbitrary topological space as the existence of a winning depth- d strategy, whether or not a least ordinal winning length exists. The finite normal form is valid in this broader sense.

An empty final state is winning. The Hider need not commit to a point in advance. For a finite decision tree, however, the game formulation and correctness on every fixed point are equivalent: a bad leaf contains two points of different colors, while an empty leaf is harmless. This equivalence will be used carefully in section 4.

Lemma 3.2 (Restriction to a subspace). *If $Z \subseteq X$ and $A \subseteq X$, then*

$$\text{sm}(A \cap Z, Z) \leq \text{sm}(A, X), \quad \text{sm}(Z) \leq \text{sm}(X).$$

Proof. Restrict every open question in a strategy on X to Z . It is open in the subspace, and every resulting state is the old state intersected with Z . For the second inequality, extend an arbitrary target of Z to a target of X . $\square$

Finite question counts are ordinary nonnegative integers. We use $\lceil \log_2 L \rceil$ only for positive integers L ; equivalently, it is the least integer d for which $L \leq 2^d$. This avoids floating-point or ordinal-logarithm conventions. In particular, $\lceil \log_2 1 \rceil = 0$.

4 Finite decision trees and alternating layers

Definition 4.1 (Alternating closed layers). For a nonempty space X and $A \subseteq X$, let $C_1 = A$ and $C_0 = X \setminus A$. A closed-layer decomposition of length $L \geq 1$ and outer color $c \in \{0, 1\}$ is a chain

$$X = F_0 \supseteq F_1 \supseteq \cdots \supseteq F_L = \emptyset \tag{4.1}$$

of closed sets such that

$$F_i \setminus F_{i+1} \subseteq C_{(c+i) \bmod 2} \quad (0 \leq i < L). \tag{4.2}$$

Empty layers are allowed. Let $L(A, X)$ be the least such length, allowing either outer color, and put $L(A, X) = \infty$ if no finite chain exists.

These are finite alternating differences of closed sets, equivalently finite Boolean combinations of open sets. This connects the game to the difference hierarchy, but the exact decision-depth statement below is proved directly.

Theorem 4.2 (Finite normal form). *For every nonempty topological space X , target A , and integer $d \geq 0$,*

$$\text{sm}(A, X) \leq d \iff L(A, X) \leq 2^d. \tag{4.3}$$

Consequently, whenever $L(A, X) < \infty$,

$$\text{sm}(A, X) = \lceil \log_2 L(A, X) \rceil. \tag{4.4}$$

4.1 From a query tree to an ordered collection of leaves

Lemma 4.3. *Complete a finite open-query tree to depth d , padding terminal branches with dummy questions. Number its leaves $0, 1, \dots, 2^d - 1$ in lexicographic order of answer strings, with $0 < 1$. If $\ell(x)$ is the leaf reached by the answers of x , then every upper level set*

$$\{x : \ell(x) \geq j\}$$

over the leaf order is open.

Proof. Proceed by induction on d . The case $d = 0$ is immediate. Let the root question be U , and put $M = 2^{d-1}$. Consider the two subtrees as query trees defined on all of X , regardless of which root answer would actually occur, and write ℓ_0, ℓ_1 for their leaf functions. If $j \leq M$, then

$$\{\ell \geq j\} = U \cup \{\ell_0 \geq j\};$$

for $j = M$, the second set is empty. If $j > M$, then

$$\{\ell \geq j\} = U \cap \{\ell_1 \geq j - M\}.$$

The induction hypothesis and closure of the topology under finite unions and intersections prove the claim. $\square$

Proof of the forward implication of theorem 4.2. A winning depth- d tree has a well-defined output color at each nonempty leaf. Assign an arbitrary color to each empty leaf. Read the 2^d leaf colors from the largest leaf down to the smallest, and group consecutive equal colors into runs. There are at most 2^d runs, and successive run colors alternate.

After deleting the largest i runs, the set of points still represented is a lower level set of ℓ . It is closed by lemma 4.3. These lower level sets, starting with X and ending with $\emptyset$, form the required chain. A run can represent no points, which is why empty layers were allowed. $\square$

4.2 Balanced search among closed layers

Proof of the reverse implication of theorem 4.2. Suppose (4.1)–(4.2) hold. Every point lies in a unique layer $F_i \setminus F_{i+1}$. The open question $X \setminus F_j$ tests whether $i < j$. Thus ordinary binary search can identify the layer index among L possibilities. Split the current interval of indices as evenly as possible at each question. Its largest remaining size after d questions is at most $\lceil L/2^d \rceil$. When $L \leq 2^d$, at most one layer remains, and that layer is monochromatic. The same reasoning applies to arbitrary Hider answers: either the state is empty or it is contained in a single layer. $\square$

Remark 4.4 (Why the logarithm is genuine). Sequentially peeling L layers would spend roughly L questions. Adaptivity permits balanced search through nested open thresholds, reducing the count to $\lceil \log_2 L \rceil$. Conversely, a depth- d tree has at most 2^d ordered leaf runs. Neither direction assumes that the space is ordered or Hausdorff.

[Added 29 September 2026, batch 52] q labels. Theorem 16.2 in Part II is the same normal form for a coloring by q labels: the least number of ordered layers with open suffixes replaces $L(A, X)$, and the depth is again its ceiling logarithm. At $q = 2$ it is theorem 4.2, proved by a second route (section 14.4).

5 Canonical alternating closures and orientation

The least layer length has a concrete canonical description. This is also the right quantity for understanding disjoint unions.

Definition 5.1 (Oriented closure sequence). For $c \in \{0, 1\}$ define

$$F_0^c = X, \quad F_{i+1}^c = \text{cl}_X(F_i^c \cap C_{(c+i+1) \bmod 2}). \quad (5.1)$$

Let $N_c(A, X)$ be the first integer L such that $F_L^c = \emptyset$, or ∞ when no such integer exists. The pair (N_0, N_1) is the *oriented closure profile* of the target.

Every F_i^c is closed and the sequence is decreasing. At step $i + 1$, one keeps the closure of points whose color would be wrong in the next layer. Therefore

$$F_i^c \setminus F_{i+1}^c \subseteq C_{(c+i) \bmod 2}.$$

Theorem 5.2 (Canonical minimality). *For each outer color c , $N_c(A, X)$ is the least possible length of a closed-layer decomposition with that color. In particular,*

$$L(A, X) = \min(N_0(A, X), N_1(A, X)). \quad (5.2)$$

Proof. If the canonical sequence terminates, its layers are a decomposition of the required length. Conversely, let $G_0, \dots, G_L$ be any closed-layer chain with outer color c . We show $F_i^c \subseteq G_i$ by induction. The assertion is true at $i = 0$. If it holds at i , then every wrong-colored point of F_i^c belongs to G_{i+1} , because $G_i \setminus G_{i+1}$ has the prescribed color. Taking closures preserves containment in the closed set G_{i+1} . Hence $F_{i+1}^c \subseteq G_{i+1}$. In particular $F_L^c = \emptyset$. $\square$

Lemma 5.3 (Two elementary profile rules). *Complementing the target swaps N_0 and N_1 . Also,*

$$N_c \leq N_{1-c} + 1$$

whenever the latter is finite.

Proof. The first assertion follows directly from (5.1). For the second, prepend an empty outer layer of color c to a decomposition whose outer color is $1 - c$; in the chain this means repeating X once at the beginning. Apply theorem 5.2. $\square$

Thus finite profile entries differ by at most one, but discarding the distinction between them can change a question count at a power-of-two threshold.

Proposition 5.4 (Exact finite profile of a sum). *Let $X = \bigsqcup_{i \in I} X_i$, $A_i = A \cap X_i$, and let $c \in \{0, 1\}$. For a nonempty family of nonempty components,*

$$N_c(A, X) = \sup_{i \in I} N_c(A_i, X_i), \quad (5.3)$$

where the supremum is taken in $\mathbb{N} \cup \{\infty\}$: an unbounded set of finite lengths has value ∞ . Hence, whenever the resulting minimum is finite,

$$\text{sm}(A, X) = \left\lceil \log_2 \left(\min_{c \in \{0, 1\}} \sup_i N_c(A_i, X_i) \right) \right\rceil. \quad (5.4)$$

Proof. Closure in a disjoint union is computed componentwise. Inductively, $F_n^c(A, X) \cap X_i = F_n^c(A_i, X_i)$ for every finite n . The global sequence is empty exactly when every component sequence is empty. Now use theorems 4.2 and 5.2. $\square$

The obstruction to (1.1) is already visible in the elementary inequality

$$\min_c \sup_i N_{i,c} \geq \sup_i \min_c N_{i,c}.$$

The two operations need not commute: each component may have its own preferred outer color, whereas one global closed-layer chain has only one outer color.

6 Exact classification at finite Cantor–Bendixson height

For a closed subspace F of a Hausdorff space, let F' be its set of nonisolated points. Set $X^{(0)} = X$, $X^{(n+1)} = (X^{(n)})'$; at a limit ordinal, use the intersection of the previous derivatives. Only finite derivatives are needed for the principal formula.

Definition 6.1. A nonempty Hausdorff space has *last derivative rank* h if

$$X^{(h)} \neq \emptyset, \quad X^{(h+1)} = \emptyset.$$

Its Cantor–Bendixson height is then $h + 1$. The terminal derivative $X^{(h)}$ is discrete; it need not be finite unless the space is compact.

Theorem 6.2 (Finite-height classification). *Let X be a nonempty Hausdorff space with last derivative rank $h < \omega$. Then*

$$\text{sm}(X) = \begin{cases} \lceil \log_2(h+1) \rceil, & |X^{(h)}| = 1, \\ \lceil \log_2(h+2) \rceil, & |X^{(h)}| \geq 2. \end{cases} \quad (6.1)$$

More precisely, the maximum of $L(A, X)$ over all targets is $h + 1$ in the first case and $h + 2$ in the second. The maxima are attained.

6.1 Why canonical closures lose one derivative at each step

Lemma 6.3. *If $F = \text{cl}_X B$ and x is isolated in F , then $x \in B$.*

Proof. An open neighborhood U with $U \cap F = \{x\}$ must meet B , because $x \in \text{cl}_X B$. Thus $x \in B$. $\square$

Lemma 6.4 (Derivative descent). *For every target and outer color, the canonical sequence satisfies*

$$F_{n+1}^c \subseteq (F_n^c)' \quad (n \geq 1), \quad F_n^c \subseteq X^{(n-1)} \quad (n \geq 1). \quad (6.2)$$

Proof. By lemma 6.3, every isolated point of F_n^c has color $(c + n) \bmod 2$. Therefore the opposite-colored points whose closure defines F_{n+1}^c all lie in $(F_n^c)'$. This derivative is closed in X , so their closure lies there too. Derivation is monotone on closed sets, giving the second assertion by induction from $F_1^c \subseteq X$. $\square$

Upper bounds in theorem 6.2. For every target and orientation, lemma 6.4 gives $F_{h+2}^c \subseteq X^{(h+1)} = \emptyset$. Hence $L(A, X) \leq h + 2$.

Now assume $X^{(h)} = \{p\}$, and write d for the target color of p . Choose c so that $(c + h) \bmod 2 = d$. We have $F_{h+1}^c \subseteq \{p\}$. If this set were nonempty, p would be isolated in it and, by lemma 6.3, would have color $(c + h + 1) \bmod 2 = 1 - d$, a contradiction. Thus $F_{h+1}^c = \emptyset$ and $L(A, X) \leq h + 1$. $\square$

6.2 Alternating rank layers give matching lower bounds

Write

$$D_j = X^{(j)} \setminus X^{(j+1)} \quad (0 \leq j \leq h).$$

These sets partition X . The set D_j is dense in $X^{(j)}$. Here are details that do not assume compactness: a finite-height space is scattered, because any nonempty subspace contains a point of least ambient rank, and that point is isolated in the subspace after restricting an isolating neighborhood at its rank. Every nonempty open subset of a scattered space contains a point isolated in that space. Apply this to $X^{(j)}$.

Lemma 6.5 (Parity target). *Color every point of D_j by $(b + j) \bmod 2$, where $b \in \{0, 1\}$, and let A_b be the color-1 set. Then*

$$N_b(A_b, X) = h + 1, \quad N_{1-b}(A_b, X) = h + 2. \quad (6.3)$$

Proof. For $h = 0$, the target is constant of color b , and the two lengths are 1 and 2, as asserted. Assume $h \geq 1$. With outer color b , the first closure is X' , because the opposite color contains D_1 , which is dense in X' , and is contained in X' . Repeating the argument shows $F_n^b = X^{(n)}$ for $0 \leq n \leq h + 1$. Thus the first empty term has index $h + 1$.

With outer color $1 - b$, the first closure is X , since it includes the dense rank-zero layer D_0 . Subsequent closures give $F_n^{1-b} = X^{(n-1)}$ for $1 \leq n \leq h + 2$. The first empty term now has index $h + 2$. The same computation covers $h = 0$. $\square$

Lemma 6.6 (Restriction of profiles). *For any subspace $Z \subseteq X$ and either color c ,*

$$N_c(A \cap Z, Z) \leq N_c(A, X).$$

Proof. The closure identity $\text{cl}_Z(B \cap Z) \subseteq Z \cap \text{cl}_X B$ gives, inductively, $F_n^c(A \cap Z, Z) \subseteq Z \cap F_n^c(A, X)$. $\square$

Lower bounds in theorem 6.2. The parity target of lemma 6.5 always has $L = h + 1$, proving the required lower bound when the terminal derivative is a singleton.

Suppose instead that $p \neq q$ lie in $X^{(h)}$. Take disjoint open neighborhoods U, V of p, q . Derivatives commute with restriction to an open subspace: $U^{(j)} = U \cap X^{(j)}$, and similarly for V . In particular both U and V have last derivative rank h .

On U , choose the parity target with bottom color 0. On V , choose the parity target with bottom color 1. Define the target arbitrarily outside $U \cup V$. By lemma 6.5, orientation 0 requires $h + 2$ layers on V , while orientation 1 requires $h + 2$ layers on U . By lemma 6.6, both global orientations require at least $h + 2$ layers. The upper bound already proved gives equality. Applying theorem 4.2 finishes the proof. $\square$

Remark 6.7 (What information survives). At finite height, the invariant sees the last derivative rank and whether the terminal derivative has one point or more than one. It sees neither the exact larger cardinality of that derivative nor the cardinalities of lower levels. Compactness and metrizability are not needed for [theorem 6.2](#).

7 Exact values on ordinal intervals

For $h, m \geq 1$ finite, put

$$X_{h,m} = [0, \omega^h \cdot m].$$

These are countable compact ordinal spaces. Standard background on ordinal spaces and Cantor–Bendixson characteristics is discussed in [\[3\]](#); the specific derivative calculation needed here is elementary.

Lemma 7.1. *For $1 \leq j \leq h$,*

$$X_{h,m}^{(j)} = \{\omega^j \eta : 0 < \eta \leq \omega^{h-j} \cdot m\}. \quad (7.1)$$

In particular,

$$X_{h,m}^{(h)} = \{\omega^h, \omega^h \cdot 2, \dots, \omega^h \cdot m\}, \quad X_{h,m}^{(h+1)} = \emptyset.$$

Proof. The nonisolated points of an ordinal interval are exactly its nonzero limit ordinals. Below $\omega^h \cdot m$, these are precisely the positive multiples of ω . This gives the formula for $j = 1$.

Within the ordered set of positive multiples of ω^j , a point with successor coefficient is isolated, including the first coefficient. A point with limit coefficient is a limit of smaller such multiples. The nonzero limit coefficients are the multiples of ω . Therefore the next derivative consists exactly of the positive multiples of ω^{j+1} in the indicated range. This proves the induction. At step h the remaining finite set is discrete, so the next derivative is empty. $\square$

Corollary 7.2 (Finite-exponent ordinal formula). *For positive finite h, m ,*

$$\text{sm}([0, \omega^h \cdot m]) = \begin{cases} \lceil \log_2(h+1) \rceil, & m = 1, \\ \lceil \log_2(h+2) \rceil, & m \geq 2. \end{cases} \quad (7.2)$$

Last rank h	One terminal point	At least two	Duplication increases sm
0	0	1	yes
1	1	2	yes
2	2	2	no
3	2	3	yes
4	3	3	no
5	3	3	no
6	3	3	no
7	3	4	yes
8	4	4	no
15	4	5	yes
31	5	6	yes
63	6	7	yes

The row $h = 0$ refers to arbitrary discrete spaces, not the notation $[0, \omega^0 \cdot m]$. A singleton has value 0; any larger discrete space has value 1.

Explicit hard targets. On $X_{h,1}$, color a point according to the parity of its Cantor–Bendixson rank. In Cantor normal form, the rank of a nonzero point is its last nonzero exponent; the point 0 has rank 0. On two copies, reverse all colors in one copy. The canonical closure lengths are then $h + 2$ in both orientations. This is an explicit witness, not merely an existence argument using the supremum over targets.

Why two copies form another ordinal space. For $h \geq 1$, the intervals

$$[0, \omega^h], \quad (\omega^h, \omega^h \cdot 2]$$

are clopen in $[0, \omega^h \cdot 2]$ and are each homeomorphic to $[0, \omega^h]$. The second interval has the same order type because removing the first point from an infinite indecomposable ordinal ω^h does not change that order type. Thus

$$X_{h,1} \sqcup X_{h,1} \cong X_{h,2}.$$

8 Sharp failures at every finite level, and sums more generally

Theorem 8.1 (Sharpness at every positive finite level). *For every integer $k \geq 1$, let*

$$K_k = [0, \omega^{2^k-1}].$$

Then

$$\text{sm}(K_k) = k, \quad \text{sm}\left(\bigsqcup_{i < \kappa} K_k\right) = k + 1 \quad \text{for every cardinal } \kappa \geq 2. \quad (8.1)$$

Thus the one-extra-move upper bound for sums is attained at every positive finite level, including by sums of two compact metrizable ordinal spaces.

Proof. Put $h = 2^k - 1$. Derivatives of a disjoint union are computed componentwise. Each component has last rank h and one terminal point; a sum of at least two has the same last rank and at least two terminal points. By theorem 6.2, the respective values are

$$\lceil \log_2(2^k) \rceil = k, \quad \lceil \log_2(2^k + 1) \rceil = k + 1.$$

□

Corollary 8.2 (Exact duplication threshold). *If X is Hausdorff of last finite derivative rank h and $|X^{(h)}| = 1$, then*

$$\text{sm}(X \sqcup X) > \text{sm}(X) \iff h + 1 \text{ is a power of 2.}$$

In that case the increase is exactly one.

Proof. The comparison is between $\lceil \log_2(h + 1) \rceil$ and $\lceil \log_2(h + 2) \rceil$. Consecutive positive integers cross a ceiling-logarithm threshold exactly when the smaller is a power of two. □

[Added 29 September 2026, batch 52] **More than two labels.** With q output labels the one-extra-move bound fails for fixed colorings: components of depth at most d can need exactly $d + \lceil \log_2 q \rceil$ questions jointly, for fixed colorings of finite sums of compact countable metrizable spaces (theorem 21.3); for $q = 3$ and $d = 1$ already of convergent sequences (section 21.1). For the supremum over colorings of finite sums of the towers $K_m \cong [0, \omega^{m-1}]$ ($m \geq 2$) the increase is still at most one (theorem 20.3 and the remark after it); in Part II's notation $K_k \cong K_{2^k}$.

8.1 Bounded finite heights

Let $\{X_i : i \in I\}$ be any nonempty family of nonempty Hausdorff spaces with finite last derivative ranks h_i . Suppose $H = \sup_i h_i < \omega$. Then H is attained and

$$\left(\bigsqcup_i X_i \right)^{(H)} = \bigsqcup_{h_i=H} X_i^{(H)}.$$

If there is a unique component of height $H + 1$ and that component has a singleton terminal derivative, the sum has value $\lceil \log_2(H + 1) \rceil$. In all other cases its value is $\lceil \log_2(H + 2) \rceil$. These two formulae can give the same numerical value.

8.2 Unbounded finite heights

The next upper bound is useful because the number of components need not be countable.

Proposition 8.3 (Parallel play with finite-time certification). *If every component X_i has finite $\text{sm}(X_i)$, then*

$$\text{sm}\left(\bigsqcup_i X_i\right) \leq \omega.$$

If the component values are unbounded among the integers, equality holds.

Proof. Fix a target A in the sum and, for each i , a finite winning strategy for $A_i = A \cap X_i$. We describe countably many rounds, each using three questions.

In round n , first ask the union of the next local questions in every component whose local strategy has not finished. Use a dummy open question in components that have finished. The single global answer advances all unfinished local simulations along the same answer history.

Next ask the union of the whole components whose current local simulation has finished with output color 0. Finally ask the analogous union for color 1. These are open sets because whole components are open. At either of these certification questions, a yes answer leaves a monochromatic state, and the Seeker has won. Empty local states may be assigned either output color.

Suppose all certification answers are no. Any point surviving the final intersection would lie in a particular component X_i . Its local strategy finishes after finitely many rounds. In that round or an earlier one, a certification question includes the whole component X_i , and a no answer removes the point. This is a contradiction. Thus the final state is empty.

The three questions per round have total order type $3 \cdot \omega = \omega$. For the lower bound when local values are unbounded, use subspace monotonicity. $\square$

In particular, for any family of finite-height Hausdorff spaces whose last ranks are unbounded in ω , the sum has membership number exactly ω . This conclusion is independent of the cardinality of the index set.

9 Finite products and the information lost by sm

Lemma 9.1 (Product derivatives). *Let X, Y be Hausdorff spaces of finite last derivative ranks h, k . For finite n ,*

$$(X \times Y)^{(n)} = \bigcup_{i+j=n} X^{(i)} \times Y^{(j)}. \quad (9.1)$$

Thus the last rank of the product is $h + k$, and its terminal derivative is $X^{(h)} \times Y^{(k)}$.

Proof. For closed subspaces $F \subseteq X$, $G \subseteq Y$, a point of $F \times G$ is isolated precisely when both coordinates are isolated in their respective subspaces. Therefore

$$(F \times G)' = (F' \times G) \cup (F \times G').$$

Derivation distributes over finite unions of closed sets: if a point is isolated or absent in every one of finitely many closed sets, intersect finitely many suitable neighborhoods to witness that it is not an accumulation point of their union. The reverse inclusion is immediate. Induction proves (9.1). At $n = h + k$ only the term with $i = h, j = k$ can be nonempty, and all terms vanish at the next stage. $\square$

Corollary 9.2 (Exact product formula). *For nonempty finite-height Hausdorff spaces $X_1, \dots, X_r$, put $H = \sum_j h_j$. Then*

$$\text{sm}\left(\prod_{j=1}^r X_j\right) = \begin{cases} \lceil \log_2(H+1) \rceil, & \text{every } X_j^{(h_j)} \text{ is a singleton,} \\ \lceil \log_2(H+2) \rceil, & \text{otherwise.} \end{cases}$$

In particular, for finite $r \geq 1$,

$$\text{sm}([0, \omega]^r) = \lceil \log_2(r+1) \rceil. \quad (9.2)$$

Knowing only the values of sm is not sufficient to compute product values. For example, let

$$P = [0, \omega^2], \quad Q = [0, \omega^3], \quad S = [0, \omega].$$

Then $\text{sm}(P) = \text{sm}(Q) = 2$, but

$$\text{sm}(P \times S) = 2, \quad \text{sm}(Q \times S) = 3.$$

The finer data consisting of the last rank and singleton-versus-multiple terminal derivative are compositional for finite products and bounded-height sums. The logarithmic compression to sm discards precisely the numerical information needed in this example.

10 Countable spaces and the transfinite-height boundary

The finite formula is not a transfinite ordinal-logarithm formula. On countable scattered spaces, all infinite Cantor–Bendixson heights lead to the same game value, namely ω .

The point-separating number $\text{ps}(X)$ is defined by the same game, but with winning condition $|W_\beta| \leq 1$ instead of monochromaticity for a target $[1]$.

Lemma 10.1 (A countable T_1 upper bound). *If X is countable and T_1 , then $\text{sm}(X) \leq \omega$. If X is also infinite, then $\text{ps}(X) = \omega$.*

Proof. Enumerate $X = \{x_0, x_1, \dots\}$. Ask the open complements $X \setminus \{x_n\}$ successively. A no answer leaves at most the singleton $\{x_n\}$. If every answer is yes, the final state is empty. This proves both upper bounds. A finite binary tree has finitely many leaves and cannot separate all points of an infinite space, proving the point-separation lower bound. $\square$

Theorem 10.2 (Countable scattered classification). *Let X be a nonempty countable scattered Hausdorff space. If its Cantor–Bendixson height is finite, theorem 6.2 applies. Otherwise*

$$\text{sm}(X) = \omega. \quad (10.1)$$

Proof. Only the lower bound in the infinite-height case remains. For every finite h , $X^{(h)}$ is nonempty and scattered, hence has an isolated point p . Choose an open neighborhood U in X with $U \cap X^{(h)} = \{p\}$. Derivatives commute with open restriction, so

$$U^{(h)} = \{p\}, \quad U^{(h+1)} = \emptyset.$$

By theorem 6.2 and subspace monotonicity, $\text{sm}(X) \geq \lceil \log_2(h+1) \rceil$. These integers are unbounded as h varies. Together with lemma 10.1, this yields (10.1). $\square$

Corollary 10.3 (All countable exponent intervals). *Let $0 < \alpha < \omega_1$ and $1 \leq m < \omega$. Then*

$$\text{sm}([0, \omega^\alpha \cdot m]) = \begin{cases} \lceil \log_2(h+1) \rceil, & \alpha = h < \omega, \ m = 1, \\ \lceil \log_2(h+2) \rceil, & \alpha = h < \omega, \ m \geq 2, \\ \omega, & \alpha \geq \omega. \end{cases}$$

Every one of these spaces has $\text{ps} = \omega$.

Proof. The finite-exponent cases were proved in corollary 7.2. When $\alpha \geq \omega$, the interval contains $[0, \omega^h]$ as a clopen subspace for every positive finite h . Their membership numbers are unbounded, while the whole space is countable and T_1 . Use lemma 10.1. $\square$

For example,

$$\text{sm}([0, \omega^\omega]) = \text{sm}([0, \omega^{\omega^2}]) = \text{sm}([0, \omega^{\omega^\omega}]) = \omega.$$

Thus extremely different countable order types can have the same membership number. This is a mathematical limitation of the invariant, not a limitation of the ordinal notation.

The countable scattered classification also covers every countable compact Hausdorff space. For completeness, the fact that such a space is scattered is proved in section A, rather than invoked as an unproved classification of countable compact spaces by ordinals.

11 A purely order-theoretic form for finite posets

Equip a finite poset P with the Alexandrov topology of upper sets. This is T_0 , and it is Hausdorff only when the order is an antichain. It is important not to mistake this topology for the order topology on the infinite ordinal spaces studied above.

Fix a target $A \subseteq P$ and color points by their membership in A . A strictly increasing chain is *alternating* if consecutive colors differ. Let a_b be the largest number of elements in an alternating chain whose greatest element has color b , with $a_b = 0$ if no point has that color.

Theorem 11.1 (Alternating-chain formula). *For every nonempty finite poset P and target A ,*

$$N_0(A, P) = 1 + a_1, \quad N_1(A, P) = 1 + a_0, \quad (11.1)$$

and therefore

$$\boxed{\text{sm}(A, P) = \lceil \log_2(1 + \min(a_0, a_1)) \rceil.} \quad (11.2)$$

Proof. Closure in the upper-set topology is downward closure. We claim that $F_n^c \neq \emptyset$ exactly when there is an alternating chain of length n ending in color $1 - c$.

More precisely, F_n^c is the downward closure of the least elements of all such chains. This is immediate for $n = 1$. In the next step, intersect with color $(c + n + 1) \bmod 2$ and take downward closure. A point of that color below the least element of an existing length- n alternating chain has the opposite color from that least element, so the inequality is strict and it can be prepended to the chain. Conversely, the first point of any length- $(n + 1)$ chain is produced this way. The induction follows.

The largest index of a nonempty F_n^c is consequently a_{1-c} , giving (11.1). Now apply theorems 4.2 and 5.2. $\square$

Corollary 11.2 (Quadratic-time computation). *Given the comparability relation of an n -element poset and a target, its optimal finite query depth is computable in $O(n^2)$ time after choosing a linear extension.*

Proof. Process points in a linear extension. Set

$$a(x) = 1 + \max\{a(y) : y < x, y \text{ has the opposite color to } x\},$$

where the maximum of the empty set is 0. Then $a_b = \max\{a(x) : x \text{ has color } b\}$. Each comparable pair is inspected at most once. Formula (11.2) completes the computation. $\square$

Example 11.3 (A finite orientation conflict). An alternating four-element chain has profile $(4, 5)$ or $(5, 4)$, depending on its orientation. Its target takes two questions. Two disjoint chains with opposite orientations have profile $(5, 5)$, so their combined target takes three questions. The archive contains an explicit optimal query tree for this eight-point example.

[Added 29 September 2026, batch 52] **Three labels.** Part II's six two-point chains 01, 02, 10, 12, 20, 21 (section 21.1) are the three-label analogue: each chain takes one question, their union three. On posets, the least layer count for q labels is the shortest common supersequence of the reduced chain words (theorem 17.1); its binary form (22.1) is (11.1) read from the other end of each chain, and for three labels the decision problem is NP-complete (theorem 22.4).

Remark 11.4. This order-theoretic formula is not obtained by applying the Hausdorff finite-height theorem to a non-Hausdorff space. Both results are separate consequences of the general finite normal form, and their hypotheses are kept distinct.

12 Algorithms, exact checks, and reproducibility

The proof of theorem 4.2 gives a strategy compiler. First construct the two canonical closure sequences; choose the shorter one; then binary-search its layer indices. At a node where the possible

indices are $a, \dots, b-1$, choose $j = \lfloor (a+b)/2 \rfloor$ and ask $X \setminus F_j$. A yes answer selects $a, \dots, j-1$, and a no answer selects $j, \dots, b-1$.

For a general infinite space, computing a closure is a mathematical operation, not necessarily an effective algorithm on a presentation of the space. The complexity claim in section 11 is specifically for finite posets whose comparability relation is supplied.

12.1 Independent verification paths

The program `code/verify.py` uses only the Python standard library. For each finite poset and every target, it compares three independently implemented calculations: exhaustive minimax over all upper-set questions, canonical downward-closure profiles, and the alternating-chain dynamic program. It also constructs a balanced strategy tree from the shorter closure chain and checks the color returned on every point.

The exhaustive minimax recurrence is

$$d_A(S) = \begin{cases} 0, & S \text{ is monochromatic,} \\ 1 + \min_U \max\{d_A(S \cap U), d_A(S \setminus U)\}, & \text{otherwise,} \end{cases}$$

where U ranges over upper sets splitting S nontrivially. States are encoded as bit masks, and the recursion is memoized. This recurrence does not use the closure formula being tested.

Number of points	Naturally labelled posets	Targets checked
1	1	2
2	2	8
3	7	56
4	40	640
5	357	11,424
6	4,824	308,736
Total	5,231	320,866

All checks passed. “Naturally labelled” means that every strict comparison $i <_P j$ satisfies $i < j$ as an integer. These are not counts of isomorphism classes. Every finite poset has at least one such labelling, so the search covers every isomorphism type through six points, with repetitions possible.

The included run took approximately 13 seconds in the execution environment used for this report. That observation is not a portable performance guarantee. The machine-readable result is in `data/verification.json`; the text log, count table, ordinal-value table, sharp-family table, and eight-point strategy certificate are provided alongside it.

12.2 Scope of the computational evidence

Finite upper-set topologies are not convergent sequences and do not model the Hausdorff accumulation behavior of ordinal intervals. Accordingly, the finite checks do *not* establish the infinite-space counterexample, the Cantor–Bendixson formula, or its transfinite consequences. Those results follow from the proofs in sections 2, 6 and 10. The computation instead checks the general finite game/closure/alternation mechanism in a large family where a direct independent minimax computation is possible.

To reproduce the included exhaustive run, execute

```
python3 code/verify.py --max-n 6
```

To rebuild the PDF, execute `make pdf`, or run `pdflatex` three times on `article.tex`. All inputs needed for compilation are local; no network access, external bibliography processor, or data regeneration is required.

[Added 29 September 2026, batch 52] Reruns. Without `--output`, `code/verify.py` writes into `data/` and replaces the recorded files there; pass `--output` with a scratch directory to keep them. Part II's verifier is `code/02-finite-output-verify.py` (section 25). Only Part II's verifier accepts `--brute-n`, and its `--max-n` stops at 6.

13 What is settled and what is not

The topological-sum identity (1.1) is false as stated. The counterexample can be expressed using the ordinary ordinal intervals $[0, \omega]$ and $[0, \omega \cdot 2]$, and the one-move defect is sharp at every positive finite level. The proof also yields exact computations for every Hausdorff space of finite Cantor–Bendixson height and every countable scattered Hausdorff space.

The two central distinctions are structural. First, the number of alternating layers is different from the number of adaptive questions: they are related by a ceiling logarithm. Second, the minimum over the two orientations is different from the pair of oriented lengths. Taking a disjoint union can force a common orientation and increase the minimum length by one; when that passes a power of two, it costs a full additional query.

No transfinite version of (4.4) is asserted. In particular, the symbol ∞ in the finite profile is not an ordinal rank and cannot simply be inserted into an ordinal logarithm. Nothing here determines the games at arbitrary uncountable lengths or resolves the other problems posed in [1]. Those are separate questions.

The research claim should be read at its demonstrated scale: an explicit negative solution to an explicitly stated question, together with a complete additional classification on a natural order-topological class. The proofs are intended to be independently checkable; their priority and presentation can be assessed separately from their mathematical content.

Part II. Finite-output open-query games

Sections 1–13 and Appendices A and B form the original report (Part I) of 19 September 2026, unchanged except for dated pointers to this addition, marked [Added 29 September 2026, batch 52]. Part II was added on 29 September 2026 in batch 52 of ProveIt’s incoming-report intake, from one manuscript written against the pinned state of Part I. It replaces the membership bit of Part I by a label from a finite alphabet of $q \geq 2$ letters. At $q = 2$ it reproduces Part I’s values (section 14.4).

14 Provenance, conventions and the binary case

14.1 One manuscript, one addition

Part II is batch-52 manuscript 12, *Finite-Output Open-Query Games: Sharp Synchronization and a Complexity Dichotomy*, an AI-assisted research article “prepared for Vladimir Reshetnikov” and dated 29 September 2026 (22 pages, US Letter, as delivered). It arrived as the archive `ProveIt_Finite_Output_Open_Query_Games.zip` (inner directory `finite_output_open_query_games/`) in ProveIt commit `11e1e9001` and was placed in `a701d9098`. It was written against ProveIt commit `5c695cfd` (in full `5c695cfd70a8b6c91bd5b2c1e99e3baecd5a864`), at which Part I’s `article.tex` had the Git blob `5e5ed2a806` and its `README.md` the blob `a8e024d1e3`. Both blobs are unchanged at the placement commit, so every statement the manuscript makes about “the ProveIt report” [4] refers to Part I as printed above.

Its code and data are shipped beside Part I’s files with the prefix `02-finite-output-` (Part I’s unprefix files count as 01): the verifier `code/02-finite-output-verify.py`, its Makefile, seven delivered records under `data/` (the verifier’s five outputs, its console log and the build log of the delivered PDF), and its source audit and research-status record in the report root. Its manuscript, README and PDF are not shipped and survive in the arrival commit. The report README lists every file and every name the delivered files still use.

The manuscript contributes, in the order printed here: a tree-layer normal form for q labels on arbitrary spaces (section 16); an exact shortest-common-supersequence representation for posets (section 17); compact countable metrizable realizations of every reduced word (section 18); the restricted-start universal-word lemma (section 19); the sharp finite Cantor–Bendixson bound $(q - 1)h + s$ and the exact q -label invariant of finite sums of convergent towers (section 20); sharp synchronization under topological sums, with exact extra cost $\lceil \log_2 q \rceil$ (section 21); a polynomial binary case against NP-complete ternary decision problems (section 22); a cyclic approximation within $\lceil \log_2(q - 1) \rceil$ questions and exact ideal search (section 23); finite obstructions (section 24); its finite checks (section 25); its proof-dependency audit, relation to the literature and a formalization plan (section 26); nine research questions (section 27); a conclusion; and certificate conventions (Appendix C). Every result, proof, example, remark, limitation and question of the manuscript is printed.

Where the merge had to choose.

- *Duplicated binary material.* The manuscript states that Part I’s binary counterexample, binary alternating-chain algorithm and binary Cantor–Bendixson results are not claimed anew. Its q -label normal form (theorem 16.2) contains Part I’s theorem 4.2 at $q = 2$, and its binary chain formula (proposition 22.1) is Part I’s theorem 11.1 in other notation. Both are printed in both

parts, because Part II states and proves them for q labels in its own notation: at $q = 2$ its proofs are second routes (Part II orders the nonempty leaves of an arbitrary tree, Part I pads the tree to full depth). The dictionary is [section 14.4](#).

- *One renamed symbol.* The manuscript’s convergent tower K_m is printed as K_m throughout Part II, because Part I’s K_k is a different space ([section 14.3](#)). No other symbol is renamed, and no normalization is changed.
- *Front matter.* The manuscript’s title block, abstract and status paragraph are printed in [section 14.2](#); its sections 1–14 are Sections 15–28 here, and its Appendix A is Appendix C. Its theorems keep their positions within each section: its Theorem $n.k$ is Theorem $(n + 14).k$ here. Its Research questions 1–9 keep their numbers.
- *Bibliography.* Both manuscripts cite Chiozini, Csernák and Soukup [1]; the two entries are merged, with both access dates. The entry [4] is Part I at the pin.
- *Shipped names.* Paths and commands in the manuscript’s [section 25](#) and Appendix C are rewritten to the shipped names; see the note there. The rewritten text promises no file that is not shipped.

Status (manuscript 12). The results of Part II are proved in ordinary mathematics. The modified-supersequence hardness theorem of Lagoutte and Tavenas ([theorem 22.2](#)) is an explicitly identified external input; the transfer to a power-of-two depth budget is proved here. Part II is not independently refereed and not proof-assistant verified; historical priority is not certified. It announces no named longstanding published open problem as solved and does not assert the word “breakthrough”. The literature on finite partitions, difference hierarchies, typed graph fusion and supersequences may contain related or equivalent formulations. The exact tower formula ([theorem 20.3](#)) concerns the supremum over all q -colorings of the specified spaces, whereas the compact sharp synchronization examples ([theorems 21.2 and 21.3](#)) concern particular colorings; the two must not be conflated. Finite computation does not prove the infinite-space statements, and the verifier’s six-point option was not run. Nothing in Part I or Part II is formalized in Lean or Rocq in ProveIt, and the Lean library that shares the `SetTheory/Cardinals` directory (large cardinals) contains nothing about open-query games.

14.2 The manuscript’s title block and abstract

Finite-Output Open-Query Games. Sharp synchronization and a complexity dichotomy. Topology / Order / Query complexity. Research article prepared for Vladimir Reshetnikov, 29 September 2026.

An open-query algorithm asks whether a hidden point belongs to an open set and must determine one of finitely many labels. We extend the binary membership framework explored in ProveIt to an arbitrary finite output alphabet. Every finite correct query tree has an ordered monochromatic layer representation, and the optimal worst-case depth is the ceiling of the binary logarithm of the minimum layer count. For upper-set topologies of posets, this layer count is exactly the shortest common supersequence length of the reduced color words of chains.

Our principal quantitative result is a sharp synchronization theorem. If each component admits at most m layers over $q \geq 2$ labels, their topological sum admits at most $(q - 1)m + 1$ layers, and this bound is best possible. Consequently, components of depth at most d can require exactly $d + \lceil \log_2 q \rceil$ questions after being combined. Sharpness holds not only for finite posets but also for fixed colorings of compact countable metrizable spaces: an iterated convergent-sequence construction realizes every reduced word with exactly its supersequence profile.

We also prove the sharp finite Cantor–Bendixson bound $(q - 1)h + s$, where h is the last derivative

rank and s is its number of colors, and compute the worst-case q -label query invariant of finite sums of convergent towers exactly.

The same representation exposes an algorithmic threshold. Optimal depth is polynomial-time computable for two labels, whereas its decision version is NP-complete for three labels, even on a disjoint union of chains. The hardness uses the known three-letter modified-supersequence theorem; an exact common-prefix padding argument handles the power-of-two query budget. In contrast, a polynomial-time cyclic-layer algorithm is within $\lceil \log_2(q-1) \rceil$ questions of optimal, hence within one question for three labels. We also give finite obstruction bounds, exact algorithms, complete proofs, and reproducible checks on 102,331 colored finite posets.

Status. The results are proved here in ordinary mathematics. The modified-supersequence hardness theorem is an explicitly identified external input. This article is not independently refereed or proof-assistant verified; historical priority is not certified.

14.3 Notation of Part II

Part II fixes a finite alphabet Σ of q letters and studies a *coloring* $c : X \rightarrow \Sigma$; Part I's target A is the case $\Sigma = \{0, 1\}$, $c = \mathbf{1}_A$. Several letters carry different meanings in the two parts. Each meaning below is fixed for its part; where a false reading is tempting, the table prints it beside the true one.

Symbol	Part I	Part II
K_k, K_m	$K_k = [0, \omega^{2^k-1}]$, with $\text{sm}(K_k) = k$ (theorem 8.1).	K_m (the manuscript's K_m) is the m -th convergent tower of lemma 18.1, of last derivative rank $m-1$; $\mathsf{K}_m \cong [0, \omega^{m-1}]$ for $m \geq 2$, so $K_k \cong \mathsf{K}_{2^k}$. <i>Not</i> $K_k = \mathsf{K}_k$: its value is $\text{sm}_2(\mathsf{K}_k) = \lceil \log_2 k \rceil$.
c	An outer color $c \in \{0, 1\}$ of a closed-layer chain; C_c its color class.	The coloring $c : X \rightarrow \Sigma$. Part I's outer color is the <i>last</i> letter w_m of a Part II word, not its first.
h	Last derivative rank.	Last derivative rank in section 20; but h_c in corollary 19.2 is the supremum of reduced chain-word lengths, and the column <code>h</code> of <code>data/02-finite-output-sharp_families.csv</code> is the word length m .
m	The multiplier in $X_{h,m} = [0, \omega^h \cdot m]$.	A number of layers, a word length, or the index of K_m .
q	In the lower-bound proof of theorem 6.2, a second point of $X^{(h)}$.	The alphabet size $ \Sigma $.
S, s	$S = [0, \omega]$.	$S \subseteq \Sigma$ a set of admissible first letters, $s = S $; in theorem 20.1, s is the number of colors on $X^{(h)}$.
D	$D_j = X^{(j)} \setminus X^{(j+1)}$; $d_A(S)$ the verifier's minimax.	$D(X, c)$ the optimal fixed-coloring depth; $D(A)$ the verifier's minimax on a mask A .
L	$L(A, X)$, the least closed-layer length.	$L(X, c)$, the least admissible word length; $L(X, \mathbf{1}_A) = L(A, X)$.
N	The oriented lengths N_0, N_1 .	N the total input length in the proof of theorem 22.4.
$A, \mathcal{A}$	The target.	$\mathcal{A}_{q,m}(S)$ a family of reduced words; A a point mask (section 23) or a number (remark after theorem 20.3).
sm, sm_q	$\text{sm}(X)$, an ordinal, the supremum over targets of the least winning length.	$\text{sm}_q(X)$, the supremum over q -colorings of the finite depth D , or ∞ . At $q = 2$, $\text{sm}_2(X) = \text{sm}(X)$ whenever either is finite.
∞	"No finite decomposition"; not an ordinal.	"No finite bound works"; not an ordinal rank.

Symbol	Part I	Part II
Layer order	$F_0 = X \supseteq F_1 \supseteq \dots$; layer 0 is the outer (open) one.	Positions $1, \dots, m$ from the bottom; suffixes are open. Part I's layer i is position $m - i$, and $F_i = \{\lambda \leq m - i\}$.

Part II's macros $\mathcal{W}$, $\mathcal{C}$, $\mathcal{A}$, scs , D , L , sm_q and $\preceq$ are defined as in the manuscript; its leaf order (the whole no-subtree before the whole yes-subtree) is Part I's lexicographic order with $0 < 1$.

14.4 The binary case: what Part II reproduces

Take $q = 2$, $\Sigma = \{0, 1\}$ and $c = \mathbf{1}_A$. The following checks were made when Part II was added; they are consequences of the two parts' proofs, not new theorems.

1. *Normal form.* A layering λ with word $w_1 \dots w_m$ is the closed-layer chain $F_i = \{\lambda \leq m - i\}$ of outer color w_m , so $\text{L}(X, c) = \text{L}(A, X)$, and theorem 16.2 gives (4.4). For an alternating word ending in the letter e , the elimination (16.3) computes Part I's canonical sequence (5.1) with outer color e term by term, so proposition 16.3 contains theorem 5.2.
2. *Posets.* Part I's a_b reads chains from their greatest element, Part II's r_a from their least; (22.1) and (11.1) both give $1 + \min(a_0, a_1)$ layers.
3. *Towers.* By lemma 18.1, $\sqcup_{i=1}^t K_m$ is Hausdorff with last derivative rank $h = m - 1$ and t terminal points, so theorem 6.2 gives $\lceil \log_2 m \rceil$ for $t = 1$ and $\lceil \log_2(m + 1) \rceil$ for $t \geq 2$: exactly (20.2) at $q = 2$. At $m = 2$, $t = 2$ this is the counterexample (2.1); at $m = 2^k$ it is (8.1) for finitely many copies. The formula was also compared with Part I's recorded tables: all 129 rows of `data/ordinal_values.csv` ($h = 0, \dots, 128$) and all 16 rows of `data/sharp_family.csv` agree.
4. *The identification* $K_m \cong [0, \omega^{m-1}]$ ($m \geq 2$, used only to name Part I's spaces). For $j \geq 1$, the space $[0, \omega^{j+1}]$ is the disjoint union of the clopen intervals $[0, \omega^j]$ and $(\omega^j n, \omega^j(n + 1)]$, $n \geq 1$, each homeomorphic to $[0, \omega^j]$ by the order-type argument of section 7, together with the point ω^{j+1} , whose neighborhoods contain all but finitely many of these intervals. A compact Hausdorff space is the one-point compactification of the complement of any of its points, so induction from $K_2 \cong [0, \omega]$ gives the claim.
5. *Sums.* At $q = 2$, theorems 21.2 and 21.3 give $m + 1$ layers and one extra question, as do lemma 5.3 and proposition 5.4; Part I's theorem 8.1 attains the extra question for the supremum over targets, Part II's families for fixed colorings. For $q \geq 3$ the exact overhead $\lceil \log_2 q \rceil$ exceeds one (section 21.1).
6. *Finite data.* The $q = 2$ rows of `data/02-finite-output-sharp_families.csv` with word lengths 2 and 4 are the chains 01, 10 (sum depth 2, a finite analogue of $S \sqcup S$) and 0101, 1010 (five layers, three questions), which is Part I's eight-point example (profile (5, 5), `data/strategy_certificate.json`).
7. *The Cantor–Bendixson bound.* At $q = 2$, (20.1) reads $\text{L} \leq h + s$. Part I's theorem 6.2 shows that both values of s that the terminal derivative allows are attained on *every* Hausdorff space of finite height, not only on towers; see the note after Research question 1.

15 The research question and its relationship to ProveIt

15.1 From a membership bit to a finite label

The natural object is a known labeling $c : X \rightarrow \Sigma$ of a topological space. A point $x \in X$ is hidden. An algorithm may ask any question of the form “is $x \in U$?”, where U is open. Questions may depend on previous answers. The algorithm must eventually report $c(x)$, not necessarily identify x . For two labels this is a set-membership problem. With more labels, different components may demand incompatible orders of resolving the labels.

Chiozini, Csernák, and Soukup introduced the topological cut-and-choose framework and asked whether the set-membership number of a topological sum always equals the supremum of its component values [1]. The ProveIt report *Open-query games on ordinal spaces* gives a binary counterexample and develops a finite alternating-layer normal form [4]. The present article starts from that direction but does not repeat its binary Cantor–Bendixson classification.

We address the following finite-output questions. How much extra information can synchronization across components require? Does the efficient binary optimization survive a third label? Can a finite non-Hausdorff obstruction be realized in familiar compact metrizable spaces? Which information, beyond a single optimal-depth number, composes correctly under topological sums?

The answers are the sharp bound $(q - 1)m + 1$, a binary/ternary complexity dichotomy, and the word profile introduced below. The compact realization result shows that the synchronization effect is not merely an artifact of finite specialization orders.

15.2 What is, and is not, being claimed

The repository baseline inspected here is the report in

```
SetTheory/Cardinals/docs/reports/
ordinals-and-order-types/games-on-ordinals/
open-query-membership-games/
```

at commit 5c695cfd70a8b6c91bd5b2c1e99e3baecd5a864. The README and the introductory part of its article were read, together with repository search results. This is a targeted comparison, not a proof audit of the whole ProveIt repository.

[Added 29 September 2026, batch 52] **The baseline is Part I.** That report is Part I above, and its `article.tex` and `README.md` are the same blobs at the pin and at the placement of this addition (section 14.1). The manuscript’s source audit (`02-finite-output-SOURCES.md`) says it read that README and lines 1–160 of the article.

The three-letter hardness of shortest common supersequences with no identical consecutive input letters is already known [5]. Related finite-partition and difference-hierarchy frameworks are also established subjects [6]. Thus neither multi-valued descriptive complexity nor the underlying supersequence hardness is announced as a discovery here. The mathematical contribution developed in this article is the explicit open-query representation, its sharp synchronization consequences, the compact word realization, the exact depth reduction, and the additive query guarantee. Their historical novelty requires a broader independent literature review.

All games in this article have *finite worst-case budgets*. An infinite value means that no finite bound works. It is not an ordinal rank and makes no claim about transfinite determinacy, countable

stopping times, or the full point-separating invariant of [1]. Except where expressly stated, we study a *fixed coloring*, not a supremum over all colorings of its space.

16 A normal form for arbitrary topological spaces

Fix a nonempty space X and a finite alphabet Σ , of size $q \geq 1$. No separation axiom is assumed.

Definition 16.1 (Depth, layers, and admissible words). For $c : X \rightarrow \Sigma$, let $D(X, c)$ be the minimum worst-case number of open questions in a finite correct deterministic decision tree, or ∞ if none exists. A word $w = w_1 \cdots w_m \in \Sigma^m$ is *admissible* if there is a map $\lambda : X \rightarrow \{1, \dots, m\}$ satisfying

$$c(x) = w_{\lambda(x)}, \quad U_j := \{x : \lambda(x) \geq j\} \text{ is open for every } j. \quad (16.1)$$

The fibers of λ are its ordered layers. Empty layers are allowed. Write $\mathcal{W}(X, c)$ for the set of admissible words, and set

$$L(X, c) = \min\{|w| : w \in \mathcal{W}(X, c)\},$$

with the minimum of the empty set interpreted as ∞ .

In a minimal representation there are no empty layers. Also, two consecutive layers carrying the same color can be merged: the remaining suffixes are still open. Thus a shortest admissible word is *reduced*, meaning that adjacent letters are distinct. This reduction is not deletion of all repeated letters; for example, 010 is reduced.

Theorem 16.2 (Tree-layer normal form). *For every (X, c) , the minimum number of nonempty leaves of a finite correct open-query tree equals $L(X, c)$. When this number is finite,*

$$D(X, c) = \lceil \log_2 L(X, c) \rceil. \quad (16.2)$$

In particular, a finite correct tree exists if and only if an admissible word exists.

Proof. Order the leaves of a query tree recursively, placing the entire no-subtree before the entire yes-subtree. We show that the union of any suffix of its ordered leaf cells is open. The assertion is proved by induction on the tree, with the current possible-point set given its subspace topology.

At the root, let the queried open set be U . If a suffix is contained entirely in the yes-subtree, induction expresses it as $U \cap V$ for an open set V of the current space. If the suffix includes part of the no-subtree, it contains the whole yes-subtree and has the form

$$U \cup ((X \setminus U) \cap V) = U \cup V,$$

again open. The same argument works in every current subspace. The one-leaf case is immediate. Removing empty leaf cells preserves the suffix property. Correctness makes each remaining leaf monochromatic. Its leaf index therefore defines a layering, so $L(X, c)$ is at most the number of nonempty leaves.

Conversely, given a layering with m nonempty layers, locate its index by balanced binary search. At an interval of candidate indices, split the interval as evenly as possible and query the open suffix U_j beginning at the split. This produces a correct tree with m leaves and depth at most $\lceil \log_2 m \rceil$.

A binary tree of depth d has at most 2^d leaves, so the first direction gives $L(X, c) \leq 2^d$ for any depth- d correct tree. The balanced construction supplies the opposite depth inequality and proves (16.2). $\square$

The theorem does not say that every optimal-depth tree has minimum leaf count. It says that minimum leaf count contains exactly the information needed to compute optimal worst-case depth. The class of all open sets is important: the proof uses closure under unions and finite intersections.

16.1 A fixed-word closure test

The following supplies a topological test without reference to a query tree. For a proposed word $w = w_1 \cdots w_m$, define closed sets by

$$G_0 = X, \quad G_{t+1} = \text{cl}_X(G_t \setminus c^{-1}(w_{m-t})) \quad (0 \leq t < m). \quad (16.3)$$

Thus we try to remove the top color, then the next color, retaining all limit points that cannot yet be discarded.

Proposition 16.3 (Greedy closed-set elimination). *The word w is admissible if and only if $G_m = \emptyset$.*

Proof. Suppose λ witnesses admissibility. Put $R_t = \{x : \lambda(x) \leq m - t\}$, so R_t is closed. Inductively $G_t \subseteq R_t$: after excluding color w_{m-t} , all remaining points of R_t lie in the closed set R_{t+1} . Taking closure therefore still stays in R_{t+1} . Consequently $G_m \subseteq R_m = \emptyset$.

Conversely, suppose $G_m = \emptyset$. Give a point removed at step $t + 1$ the index $m - t$. The removed set $G_t \setminus G_{t+1}$ has color w_{m-t} because G_{t+1} contains every point of G_t of another color. These removed sets partition X . The suffix beginning at index j is $X \setminus G_{m-j+1}$, which is open. This is the required layering. $\square$

The proposition is a finite colored elimination statement. It should not be confused with a classification of all transfinite partition hierarchies.

17 Posets and shortest common supersequences

Give a poset P its upper Alexandrov topology: U is open exactly when $x \in U$ and $x \leq y$ imply $y \in U$. A map $\lambda : P \rightarrow [m]$ has open suffixes exactly when it is order-preserving. This observation turns topology into a word-constrained monotone-map problem.

For a finite nonempty chain $x_1 < \cdots < x_r$, read the word $c(x_1) \cdots c(x_r)$ and collapse each maximal run of equal letters to one letter. Let $\mathcal{C}(P, c)$ denote the collection of all resulting reduced words. Write $u \preceq v$ when u is a subsequence of v , using a strictly increasing map of positions. For a word family $\mathcal{F}$, $\text{scs}(\mathcal{F})$ is the minimum length of a common supersequence, or ∞ when no finite common supersequence exists.

Theorem 17.1 (Exact chain-language representation). *For any poset, finite or infinite, and any finite word v ,*

$$v \in \mathcal{W}(P, c) \iff u \preceq v \text{ for every } u \in \mathcal{C}(P, c). \quad (17.1)$$

Consequently

$$L(P, c) = \text{scs}(\mathcal{C}(P, c)), \quad D(P, c) = \lceil \log_2 \text{scs}(\mathcal{C}(P, c)) \rceil \quad (17.2)$$

whenever the right-hand side is finite.

Proof. An admissible map is weakly increasing along a chain. At a color change its index must increase strictly, since one word position has only one color. Selecting one point per color run embeds the reduced chain word into v .

For the converse, greedily embed each reduced finite chain word into v . For $x \in P$, consider all finite chains ending at x , including the singleton chain. Their greedy last positions belong to the finite set $\{j : v_j = c(x)\}$. They are nonempty in number and all exist by hypothesis. Let $\lambda(x)$ be their maximum.

If $x < y$, take a chain ending at x whose greedy last position is $\lambda(x)$ and append y . If the last color repeats, reducing the word leaves the last greedy position unchanged; otherwise that position moves forward. The extended chain ends at y , so $\lambda(y) \geq \lambda(x)$. Also $v_{\lambda(x)} = c(x)$. Thus λ is a color-respecting order-preserving map, which proves admissibility. The formulas follow from Theorem 16.2. $\square$

17.1 A linear-time feasibility test on a finite DAG

Let the order be the reachability relation of an acyclic directed graph with n vertices and e edges, supplied in a topological order. The input may contain just cover edges, all strict comparisons, or any generating DAG. For a finite word v , write

$$\text{next}_a(b) = \min\{j \geq b : v_j = a\},$$

where the minimum is ∞ if the set is empty. Process the vertices in order and compute

$$b(x) = \max(\{1\} \cup \{f(y) : y \in \text{Pred}(x)\}), \quad f(x) = \text{next}_{c(x)}(b(x)). \quad (17.3)$$

Here $\text{Pred}(x)$ consists of incoming neighbors, not necessarily all predecessors. If a value is infinite the test rejects.

Proposition 17.2 (Least feasible map). *If the test succeeds, f is the pointwise least order-preserving map compatible with v and c . It succeeds exactly when v is admissible. With a next-position table, its running time is $O(q|v| + n + e)$.*

Proof. If g is any compatible monotone map, induction gives $f(y) \leq g(y)$ at already processed vertices. Hence $b(x) \leq g(x)$, and $g(x)$ is an occurrence of $c(x)$. The first possible such occurrence is at most $g(x)$. This proves minimality and shows that rejection precludes any compatible map. When all values are finite, monotonicity holds on every edge and hence on reachability. The recurrence uses one scan of the graph after constructing the next-position table backwards through the word. $\square$

The weak inequality $j \geq b(x)$ is essential. Comparable points of the same color may use the same layer. Replacing it by $j > b(x)$ would compute a different problem and would make even a constant-colored long chain unnecessarily hard.

Corollary 17.3 (Disjoint chains). *If P is a disjoint union of finite chains whose reduced color words are $w_1, \dots, w_r$, then*

$$L(P, c) = \text{scs}(w_1, \dots, w_r).$$

Proof. Every chain lies in a single component. Its reduced word is a subsequence of the component's reduced word. Thus the component words already impose all the constraints in Theorem 17.1. $\square$

18 Compact metrizable realizations of word profiles

Finite upper-set spaces are generally not Hausdorff. We now show that their word obstructions can survive in compact countable metrizable spaces. The result concerns the entire admissible-word profile, not only one numerical invariant.

Define K_1 to be a singleton. Recursively, K_m is the one-point compactification of the topological sum of countably many copies of K_{m-1} . Denote the new point by ∞_m . In particular, K_2 is a convergent sequence with its limit.

Lemma 18.1 (The convergent towers). *Each K_m is countable, compact, and metrizable. Its last nonempty Cantor–Bendixson derivative is $K_m^{(m-1)} = \{\infty_m\}$ for $m \geq 2$; $K_1^{(0)} = K_1$ is also a singleton. A neighborhood of ∞_m contains all but finitely many entire copies of K_{m-1} .*

Proof. A compact subset of a topological sum of compact components meets only finitely many components: the components themselves form an open cover. The assertion about neighborhoods follows directly from the one-point compactification. Countability and compactness follow by induction and this neighborhood property.

For metrizability, choose a metric $d_{m-1} \leq 1$ on each copy. Label copies by positive integers n . Within copy n , use $2^{-n}d_{m-1}$; give each of its points distance 2^{-n} from ∞_m ; and give points in distinct copies n, k distance $2^{-n} + 2^{-k}$. This is a metric inducing precisely the stated topology. Each copy is open and closed. Taking derivatives inside a copy agrees with its own derivatives, while ∞_m remains a limit point as long as infinitely many nonempty copies remain. Induction gives the derivative formula. $\square$

For a reduced word $w = a_1 \cdots a_m$, color K_m recursively: give ∞_m color a_1 , and give every copy of K_{m-1} the coloring for $a_2 \cdots a_m$. For $m = 1$, color its single point a_1 . Call this coloring c_w . Equivalently, a point of Cantor–Bendixson rank r has color a_{m-r} .

Theorem 18.2 (Exact compact word realization). *For every reduced word w and every finite word v ,*

$$v \in \mathcal{W}(K_{|w|}, c_w) \iff w \preceq v. \quad (18.1)$$

In particular, $L(K_m, c_w) = m$ and $D(K_m, c_w) = \lceil \log_2 m \rceil$.

Proof. The word w itself is admissible. Put the top point in layer 1, and use the recursive suffix layering, shifted by one, in every component. Suffixes starting after layer 1 are unions of open subsets of the open components; the first suffix is the whole space. Embedding these layer indices in any supersequence v preserves the openness of suffixes, so $w \preceq v$ implies admissibility.

For the converse, induct on m . The singleton case is immediate. Suppose λ witnesses admissibility of v on K_m , and set $j = \lambda(\infty_m)$, so $v_j = a_1$. The open suffix $\{x : \lambda(x) \geq j\}$ contains ∞_m . By Lemma 18.1, it contains an entire copy of K_{m-1} . On that copy the restricted layering uses only indices $j, \dots, |v|$. The induction hypothesis embeds $a_2 \cdots a_m$ in $v_j \cdots v_{|v|}$. Since $a_2 \neq a_1 = v_j$, the first position of this embedding is strictly greater than j . Prefixing position j embeds $a_1 \cdots a_m = w$ into v , as required. $\square$

Corollary 18.3 (Compact realization of any finite colored poset). *Every finite colored poset has the same admissible-word profile as some compact countable metrizable colored space of finite Cantor–Bendixson height.*

Proof. There are finitely many reduced chain words. Take the finite topological sum of one compact realization for each such word. A word is admissible on this sum exactly when it is admissible on every component: restriction proves necessity, and componentwise gluing proves sufficiency. Theorem 18.2 and Theorem 17.1 now give equality of profiles. A finite sum of compact countable metrizable spaces has the same properties. $\square$

This construction is an existence theorem and can be exponentially large if one lists all chain words of a poset. For a given list of chain words, however, it preserves that list directly as a succinct description of the compact instance.

19 The extremal supersequence lemma

The exact constants come from a small extremal word problem. Let $\mathcal{A}_{q,m}(S)$ consist of all reduced words of length $m \geq 1$ over Σ whose first letter lies in a fixed nonempty set $S \subseteq \Sigma$. Write $s = |S|$ and $\mathcal{A}_{q,m} = \mathcal{A}_{q,m}(\Sigma)$.

Theorem 19.1 (Restricted-start universal words). *For $q \geq 2$ and $1 \leq s \leq q$,*

$$\text{scs}(\mathcal{A}_{q,m}(S)) = s + (q - 1)(m - 1). \quad (19.1)$$

In particular,

$$\text{scs}(\mathcal{A}_{q,m}) = (q - 1)m + 1. \quad (19.2)$$

Proof. Order the alphabet cyclically, with the s letters of S first, and repeat this order indefinitely. Greedily embedding a word in $\mathcal{A}_{q,m}(S)$ puts its first letter at a position at most s . Every next letter differs from its predecessor and therefore occurs within the next $q - 1$ positions. The prefix of length $s + (q - 1)(m - 1)$ is thus a common supersequence.

For the lower bound, let v be any common supersequence. Choose a first letter in S whose first occurrence is latest. Its position is at least s . Having chosen a prefix ending with letter a at its greedy position j , consider the next occurrences after j of all $q - 1$ letters different from a . Every such occurrence exists while the chosen prefix has length less than m : otherwise that prefix followed by the missing letter could be extended to a word in $\mathcal{A}_{q,m}(S)$ that does not embed in v .

These $q - 1$ next occurrences occupy different positions. Choose the latest; its distance from j is at least $q - 1$. Continuing this adversarial choice gives a reduced word of length m whose greedy last position is at least $s + (q - 1)(m - 1)$. Its existence forces the same lower bound on $|v|$. $\square$

Corollary 19.2 (Bounded color alternation is exactly finite solvability). *For a nonempty colored poset, let h_c be the supremum of lengths of reduced finite chain words. A finite correct open-query tree exists if and only if $h_c < \infty$. In that case*

$$h_c \leq L(P, c) \leq (q - 1)h_c + 1 \quad (q \geq 2). \quad (19.3)$$

The upper bound is sharp for every $q \geq 2$ and $h_c \geq 1$.

Proof. Every chain word must embed into an admissible word, giving the lower bound. If all reduced chain words have length at most h_c , extend each shorter one to a reduced word of length h_c . The universal word from Theorem 19.1 embeds all of them, so Theorem 17.1 gives the upper bound. A disjoint union of chains carrying every word of $\mathcal{A}_{q,h_c}$ attains equality. Unbounded lengths preclude any finite word. $\square$

The lower bound in (19.3) can be supplemented by the number of colors actually present. Neither bound alone determines the answer: chain words of the same length may synchronize in very different ways.

20 Sharp Cantor–Bendixson bounds and exact tower invariants

For a space X , let $X^{(0)} = X$ and let $X^{(r+1)}$ be the nonisolated-point subspace of $X^{(r)}$. Suppose $X^{(h)} \neq \emptyset$ and $X^{(h+1)} = \emptyset$, where h is a nonnegative integer. The Cantor–Bendixson rank of x is the unique r with $x \in X^{(r)} \setminus X^{(r+1)}$.

Theorem 20.1 (A sharp finite-height layer bound). *Let X be a nonempty Hausdorff space of last derivative rank $h < \infty$. Let $c : X \rightarrow \Sigma$, where $q = |\Sigma| \geq 2$, and let $s = |c(X^{(h)})|$ be the number of colors appearing on its last nonempty derivative. Then*

$$L(X, c) \leq (q - 1)h + s, \quad D(X, c) \leq \lceil \log_2((q - 1)h + s) \rceil. \quad (20.1)$$

For every $q \geq 2$, $h \geq 0$, and $1 \leq s \leq q$, the layer bound is attained by a compact countable metrizable colored space having exactly s points in its last nonempty derivative, with distinct colors on those points.

Proof of the upper bound. Choose a cyclic order of the alphabet whose first s entries are the colors on $X^{(h)}$. Let v be the prefix of the repeated order of length $M = s + h(q - 1)$. Give a point of rank h the first occurrence of its color in v ; its index is at most s .

For a point of rank $r < h$, put

$$b_r = s + (h - 1 - r)(q - 1)$$

and assign it the first occurrence of its color at or after b_r . The interval $[b_r, b_r + q - 1]$ has q positions and contains the whole alphabet, so this assignment exists and lies in $[1, M]$. For two consecutive lower ranks, the maximum permitted index at the higher rank equals the minimum permitted index at the lower rank. The corresponding comparison also holds between rank h and rank $h - 1$: their boundary is s .

Every point x of rank r has an open neighborhood V_x satisfying $V_x \cap X^{(r)} = \{x\}$, because it is isolated in $X^{(r)}$. Every other point of V_x has smaller rank and hence an assigned index at least that of x . Thus $V_x \subseteq \{y : \lambda(y) \geq \lambda(x)\}$. Each suffix of λ is a union of such neighborhoods, and is open. This proves admissibility of v . For $h = 0$, the same argument simply orders the s colors of the discrete space. The depth estimate follows from Theorem 16.2. $\square$

To prove sharpness we need colorings richer than a single rank word. For $a \in \Sigma$ and $m \geq 2$, color the top point of K_m by a . List every reduced word u of length $m - 1$ whose first letter differs from a , and use infinitely many component copies with coloring c_u for each listed word. There are finitely many word types and countably many copies. Denote the resulting coloring by $b_{a,m}$. Put $b_{a,1} = a$ on K_1 .

Lemma 20.2 (All words with one prescribed initial color). *For every $m \geq 1$,*

$$\mathcal{W}(K_m, b_{a,m}) = \{v : w \preceq v \text{ for every } w \in \mathcal{A}_{q,m}(\{a\})\}.$$

Proof. For $m = 1$ the assertion is the singleton case. For $m \geq 2$, suppose v is admissible and its layering gives the top point index j , with $v_j = a$. The open suffix beginning at j contains all but finitely many component copies. Since each word type u occurs infinitely often, one entire copy of each type lies in that suffix. Theorem 18.2 embeds u into $v_j \cdots v_{|v|}$, and its first letter differs from a , so this embedding starts after j . Prefixing j embeds every word au in the required family.

Conversely, suppose every au embeds into v . Let j be the first occurrence of a in v . In an embedding of any au , the initial a can be replaced by this earlier occurrence, so every u embeds after j . Use such an embedding and Theorem 18.2 to place the layers of every component after j ; place the top point at j . Suffixes beginning at most at j are the whole space, while later suffixes are unions of open subsets of components. This is an admissible layering. $\square$

Completion of the sharpness proof of Theorem 20.1. Let S be a set of s alphabet letters and take one copy of $(K_{h+1}, b_{a,h+1})$ for each $a \in S$. For $h = 0$ these are s singletons. The last derivative consists of their s top points. By Lemma 20.2, the word profile of the sum requires exactly the words of $\mathcal{A}_{q,h+1}(S)$ as subsequences. Its minimum word length is $s + (q - 1)h$ by Theorem 19.1. $\square$

Now take a supremum over all colorings, in contrast to the fixed-coloring invariant $D(X, c)$:

$$\text{sm}_q(X) := \sup_{c: X \rightarrow \Sigma} D(X, c).$$

The alphabet has q available colors; a coloring need not use all of them.

Theorem 20.3 (Exact finite-output invariant of convergent towers). *For all integers $m, t \geq 1$ and $q \geq 2$,*

$$\text{sm}_q\left(\bigsqcup_{i=1}^t K_m\right) = \lceil \log_2((q-1)(m-1) + \min(q, t)) \rceil. \quad (20.2)$$

In particular,

$$\text{sm}_q(K_m) = \lceil \log_2(1 + (q-1)(m-1)) \rceil. \quad (20.3)$$

Proof. The last nonempty derivative is at rank $m - 1$ and has t points. Every coloring uses at most $\min(q, t)$ colors there. The upper bound is therefore Theorem 20.1. For the lower bound, choose $\min(q, t)$ distinct top colors and the corresponding colorings $b_{a,m}$. If extra copies remain, duplicate any one of those colored components. Duplicating its word constraint does not change the intersection of word profiles. The sharpness construction in Theorem 20.1 gives exactly the claimed layer count and hence depth. $\square$

For $q = 2$, (20.2) is $\lceil \log_2 m \rceil$ for one tower and $\lceil \log_2(m + 1) \rceil$ for two or more, consistent with the binary repository result [4]. The theorem above supplies the finite-alphabet extension for this compact family, including its worst-case colorings and explicit proofs.

[Added 29 September 2026, batch 52] Checked against Part I. The binary result is theorem 6.2 applied to $\bigsqcup_{i=1}^t K_m$ (last derivative rank $m - 1$, t terminal points). At $m = 2$, $t = 2$ the value 2 is Part I's counterexample (2.1); since $K_{2^k} \cong K_k$, the values k and $k + 1$ at $m = 2^k$ are (8.1). The formula agrees with all rows of Part I's recorded tables (section 14.4, items 3 and 4).

Remark 20.4 (A limitation and a useful distinction). Theorem 20.1 gives a sharp *uniform bound over a class of spaces*; it does not assert that every coloring attains it, or that every Hausdorff space of the specified height has the same multi-output invariant. Theorem 20.3 is an exact statement for the

defined towers and their finite sums. When $m \geq 2$, its formula shows that summing these uncolored towers can increase sm_q by at most one, even though fixed colorings can exhibit the larger increase established next. Indeed, if $A = (q-1)(m-1)+1$, then $A \leq (q-1)(m-1)+\min(q, t) \leq A+q-1 \leq 2A$.

21 Sharp synchronization under topological sums

Proposition 21.1 (Profiles compose by intersection). *For any nonempty family of nonempty colored spaces, with the same alphabet,*

$$\mathcal{W}\left(\bigsqcup_{i \in I} X_i, \bigsqcup_{i \in I} c_i\right) = \bigcap_{i \in I} \mathcal{W}(X_i, c_i). \quad (21.1)$$

Proof. Restrict a layering of the sum to each component for one inclusion. For the other, take a layering of each component with the same word. Their maps glue, and every glued suffix is open because its restriction to each open component is open. One can choose component layerings by any such witnesses; alternatively Proposition 16.3 provides canonical layers for the prescribed word. $\square$

A scalar such as minimum depth or minimum length loses this compatibility information. Even the collection of shortest words alone need not be used as a substitute for the full admissibility profile: a longer locally admissible word may be useful for synchronization.

Theorem 21.2 (Sharp layer synchronization). *Let $q \geq 2$ and $m \geq 1$. If $L(X_i, c_i) \leq m$ for every i , then*

$$L\left(\bigsqcup_i X_i, \bigsqcup_i c_i\right) \leq (q-1)m + 1. \quad (21.2)$$

For every q, m the constant is best possible. Sharp examples can be chosen as finite disjoint unions of m -point chains, or as finite disjoint unions of copies of the compact countable metrizable space K_m with fixed rank-word colorings.

Proof. Each component admits a reduced word of length at most m . Every such word embeds into a reduced word of length m and hence into the single universal word of length $(q-1)m + 1$ in Theorem 19.1. Inserting empty layers according to this embedding makes the universal word admissible on every component. Proposition 21.1 gives the upper bound, with no restriction on the number of components.

For sharpness, take one component for each $w \in \mathcal{A}_{q,m}$, either the chain carrying w or the compact colored tower (K_m, c_w) . Each component has exactly m layers. By Corollary 17.3 or Theorem 18.2, the union's profile is exactly the common supersequence profile of $\mathcal{A}_{q,m}$, whose minimum length is $(q-1)m + 1$. There are $q(q-1)^{m-1}$ components. $\square$

Theorem 21.3 (Exact maximal extra query cost). *For $q \geq 2$ and $d \geq 0$, if every component has fixed-coloring depth at most d , then*

$$D\left(\bigsqcup_i X_i, \bigsqcup_i c_i\right) \leq d + \lceil \log_2 q \rceil. \quad (21.3)$$

Equality is attainable for every pair (q, d) , both by finite colored posets and by finite families of compact countable metrizable colored spaces.

Proof. Use Theorem 16.2 and Theorem 21.2 with $m = 2^d$. If $k = \lceil \log_2 q \rceil$, then $2^{k-1} \leq q - 1 < 2^k$, so

$$2^{d+k-1} < (q-1)2^d + 1 \leq 2^{d+k}.$$

Thus the ceiling of its logarithm is exactly $d + k$. The sharp examples of Theorem 21.2 have each component depth exactly d and union depth exactly $d + k$. $\square$

There is a simple independent intuition for the upper depth bound. Synchronize component trees for d rounds, taking at each round the union of the appropriate component queries. After that transcript, the surviving points in each component have one known color, but different components may have different colors. Queries that are unions of whole components resolve the remaining q possibilities in at most $\lceil \log_2 q \rceil$ rounds. The layer theorem does more: it finds the exact extremal number of layers and proves the depth loss sharp.

For the supremum-over-colorings invariant, the same upper estimate holds if $\text{sm}_q(X_i) \leq d$ for all i . It is sharp on finite upper-topology chains of 2^d points: their worst-case layer count is 2^d , attained by an alternating two-color word. The compact sharpness construction in Theorem 21.3 is a statement about its *specified colorings*. It does not assert that $\text{sm}_q(K_{2^d}) = d$ for $q > 2$; Equation (20.3) gives the different, correct value.

21.1 The first genuinely three-color example

Take six two-point chains with color words

$$01, \quad 02, \quad 10, \quad 12, \quad 20, \quad 21.$$

Each needs one question. The word 01201 is a common supersequence, with the following embeddings:

01	02	10	12	20	21
(1, 2)	(1, 3)	(2, 4)	(2, 3)	(3, 4)	(3, 5)

No word of length four contains all six ordered pairs. Such a word must use all three colors, and at least two of the colors occur only once; only one order of those two colors can then be a subsequence. Hence the sum has exactly five layers and exactly three questions. In particular, the binary “one extra question” bound is false for fixed three-color targets.

Replace each chain ab by a convergent sequence whose limit has color a and all isolated points have color b . Theorem 18.2 preserves every word constraint. This gives the same one-to-three increase on a compact countable metrizable space that is a finite sum of convergent sequences.

The six constraints are deletion-minimal for this three-query obstruction. More generally, if the ordered pair ba is deleted from the full q -color pair family, let R list all colors except a, b . The word $RabR$ has length $2q - 2$ and contains every remaining ordered pair. For $q = 3$ this is a four-layer, two-query solution after deleting any one component.

22 A binary/ternary complexity dichotomy

A finite input poset is given by a generating DAG or by its order relation, with an explicit color at every vertex. The decision problems ask whether $L(P, c) \leq k$ or whether $D(P, c) \leq d$, with an integer budget supplied as part of the input.

22.1 The binary case

For colors 0, 1, let r_a be the maximum length of a reduced chain word beginning with a , or zero if that color never occurs. Every reduced binary word alternates. An alternating master word starting with a embeds a reduced chain word beginning with a as soon as it has enough positions; embedding a word beginning with $1 - a$ costs one additional initial position.

Proposition 22.1 (Exact binary formula). *For a nonempty finite binary-colored poset,*

$$L(P, c) = \min\{\max(r_0, r_1 + 1), \max(r_1, r_0 + 1)\}. \quad (22.1)$$

Both L and D are polynomial-time computable.

Proof. A shortest master word is reduced and therefore alternating. The two possible initial colors give the two displayed lengths, by the preceding observation and Theorem 17.1. To compute the r_a , process the transitive order in a linear extension. For each vertex x and each starting color a , store the largest reduced-chain length beginning with a and ending at x . Extending a chain from $y < x$ increases the value by 1 precisely when $c(y) \neq c(x)$. This is a polynomial dynamic program; using an explicit order relation costs $O(n^2)$ after it is available. Finally apply Theorem 16.2. $\square$

This is the binary phenomenon already present in the repository report [4], now expressed in the chain-language notation. It is included to make the contrast with three colors precise, not as a new binary theorem.

[Added 29 September 2026, batch 52] **Where it is in Part I.** Theorem 11.1 and its quadratic-time corollary. Part I's a_b reads a chain from its greatest element, r_a here from its least; both formulas give $1 + \min(a_0, a_1)$ layers (section 14.4, item 2).

22.2 The imported hardness theorem

We use the following established result and no other external hardness assumption beyond ordinary NP-completeness terminology.

Theorem 22.2 (Lagoutte–Tavenas, modified SCS; external input). *Over three letters, deciding whether a finite family of words has a common supersequence of length at most k is NP-complete even when no input word has identical consecutive letters and no input word begins with a designated letter, which may be called 2.*

This is Corollary 6 of [5]; its strict-budget convention is equivalent by subtracting one from the budget. Its relationship to earlier typed-DAG fusion work of Darte is discussed there. We do not reprove that hardness theorem or claim that the reduced-word restriction is new.

Corollary 17.3 immediately transfers this result to the layer budget. The depth budget requires care: simply rounding $\log_2 k$ loses the exact threshold. The next lemma supplies the missing step.

Lemma 22.3 (Common-prefix cancellation). *For any finite word p and any nonempty finite family of finite words,*

$$\text{scs}(pw_1, \dots, pw_r) = |p| + \text{scs}(w_1, \dots, w_r). \quad (22.2)$$

Proof. Prefixing p to a shortest common supersequence proves the upper bound. For the reverse bound, let v contain every pw_i . Greedily embed p into v , and let its last position be j . For any

embedding of pw_i , the prefix p finishes at a position at least j , so the embedded copy of w_i lies entirely after j . Consequently the suffix of v after j is a common supersequence of all w_i . Since $j \geq |p|$, this gives the lower bound. For the empty prefix the assertion is immediate. $\square$

Theorem 22.4 (Three colors suffice for NP-completeness). *For every fixed alphabet size $q \geq 3$, each of the following is NP-complete: deciding $L(P, c) \leq k$, and deciding $D(P, c) \leq d$. The result holds already for disjoint unions of chains.*

Proof. First consider membership in NP. Every n -point colored poset has a layering of at most n positions, obtained by listing its vertices in a linear extension. A word of length at most n together with a color-respecting monotone map to its positions is a polynomial certificate. Its correctness is checked on the generating edges. For the depth problem, test whether its length is at most 2^d . If $d \geq \lceil \log_2 n \rceil$, the answer is automatically yes; otherwise the integer 2^d is at most $2n$ and causes no encoding problem. The same certificate proves membership for a layer budget.

Layer hardness follows directly from Theorem 22.2 by making a chain for each input word. For depth hardness, take a modified-SCS instance $(w_1, \dots, w_r, k)$ over $\{0, 1, 2\}$ whose words do not begin with 2. Delete empty words. Trivial budgets are decidable directly and can be mapped to fixed yes- or no-instances, so assume $1 \leq k < N$, where $N = \sum_i |w_i|$.

Set

$$d = \lceil \log_2 k \rceil, \quad B = 2^d, \quad t = B - k.$$

If $t > 0$, choose the alternating word over $\{0, 2\}$ of length t ending in 2; call it p . If $t = 0$, let p be empty. Every word pw_i is reduced, including across the joining boundary, because no w_i starts with 2. Let P' be the disjoint union of chains carrying the words pw_i . By Corollary 17.3 and Lemma 22.3,

$$L(P', c') = t + \text{scs}(w_1, \dots, w_r).$$

Therefore

$$\begin{aligned} D(P', c') \leq d &\iff L(P', c') \leq B \\ &\iff \text{scs}(w_1, \dots, w_r) \leq B - t = k. \end{aligned}$$

The output has $N + rt$ vertices. Since $t < k \leq N$, its size is polynomial in the input size. This is an exact many-one reduction for the depth threshold. The three used colors can be viewed as a subset of any larger fixed alphabet. $\square$

22.3 The same hardness on succinct compact instances

The compact realization has an algorithmic consequence if the representation of an infinite space is specified rather than left implicit. Define a *tower-list instance* to be a finite list of reduced words $w_1, \dots, w_r$, denoting

$$\bigsqcup_{i=1}^r (K_{|w_i|}, c_{w_i}).$$

Its input size is the total word length, not the cardinality of the infinite underlying space. The allowed open sets are still all open sets of that space.

Corollary 22.5 (Compact rank-word instances). *For a fixed alphabet of at least three letters, deciding an exact finite layer budget or an exact finite query-depth budget on tower-list instances is NP-complete.*

Proof. Theorem 18.2 and Proposition 21.1 identify the entire admissible-word profile with the common supersequence profile of the input list. Thus the same certificates and the same padding reduction apply. $\square$

This is a precisely defined succinct-input result, not a claim about an unspecified computational presentation of arbitrary topological spaces.

23 Efficient approximation and exact algorithms

23.1 Cyclic master words

Fix a cyclic order of the q letters. For each of its q rotations, greedily map the input poset into the corresponding infinite periodic word using (17.3). Next occurrences are computed by modular arithmetic, so no infinite object is stored. Let M_r be the largest position used for rotation r . Return the shortest of the resulting prefixes, with its greedy rank map, and perform balanced suffix queries.

Theorem 23.1 (Additive approximation in query count). *For a finite nonempty q -colored poset, $q \geq 2$, the cyclic algorithm runs in $O(q(n + e))$ word operations and returns an admissible prefix of length at most*

$$M_{\text{cyc}} \leq 1 + (q - 1)(L(P, c) - 1). \quad (23.1)$$

Its query tree has depth at most

$$D(P, c) + \lceil \log_2(q - 1) \rceil. \quad (23.2)$$

For two colors it is exact. For three colors it is within one additional question of optimal.

Proof. Let w be a shortest admissible word of length L . It is reduced. Choose the cyclic rotation starting with its first letter. Greedily embed w into that periodic word. Its first position is 1, and every successive letter advances by at most $q - 1$, giving a last position at most $1 + (q - 1)(L - 1)$. Composing this word embedding with a layering for w produces a compatible monotone map into the periodic prefix. The greedy map is pointwise least by Proposition 17.2, so its last position is no larger. Minimizing across rotations proves (23.1).

Since $M_{\text{cyc}} \leq (q - 1)L$, taking logarithms and ceilings gives

$$\lceil \log_2 M_{\text{cyc}} \rceil \leq \lceil \log_2 L \rceil + \lceil \log_2(q - 1) \rceil.$$

The normal form converts this to the depth estimate. For $q = 2$, the length bound is $M_{\text{cyc}} \leq L$, forcing equality with the optimum.

Each rotation requires one graph scan. Every rank is at most qn , since at most n successive vertex assignments can increase a rank, each by at most q . The ranks therefore occupy $O(\log(qn))$ bits. This proves the stated word-operation bound; bit-operation costs include the corresponding integer arithmetic factor. $\square$

With alphabet $\{0, \dots, q - 1\}$ and rotation starting with a , the next position of color c at or after a 1-based lower bound b is

$$b + (c - a - (b - 1) \bmod q),$$

where the remainder is in $\{0, \dots, q - 1\}$. This is the implementation used in the verifier.

Empty positions may be deleted, and adjacent same-color occupied layers may then be merged, before building a query tree. These improvements only help; the theorem does not depend on applying them. The raw prefix-length bound is sharp for this particular algorithm: a chain following the reverse cyclic order has its first match at position 1 in the best rotation and each later match $q - 1$ positions farther along. For example, the chain 2102 has optimum length 4 but uses positions 1, 3, 5, 7 in the best 012-cycle rotation. Deleting its unused positions recovers the optimum, so this example is *not* a lower bound against improved approximation algorithms.

23.2 Exact search on ideals

Let I be an order ideal, interpreted as the points already placed into early layers. Reading color a can process all points whose as-yet-unprocessed predecessors have that color. More precisely, define

$$T_a(I) = I \cup \{x \in P : \downarrow x \setminus I \subseteq c^{-1}(a)\}. \quad (23.3)$$

It is an ideal. It can be computed by a topological scan that adds a vertex of color a as soon as all its predecessors have been added.

Proposition 23.2 (Exact ideal automaton). *Starting at $\emptyset$, a word is admissible exactly when the successive transitions T_a reach P . Breadth-first search of this finite state graph therefore returns a shortest admissible word. If $J(P)$ is the number of order ideals, the running time is $O(qJ(P)(n + e))$, with at most $J(P) \leq 2^n$ states.*

Proof. After a word prefix has been read, process every vertex as early as its color and predecessors allow. Equation (23.3) is precisely the maximal possible extension using the next single color: a point can be included if and only if every still-unprocessed point below it can be included in this same monochromatic layer. Induction on the word length shows that this process contains the processed ideal of any other compatible partial layering. Thus it reaches all vertices if and only if some complete layering exists. Breadth-first search minimizes the number of transitions. Each of the q transitions per reached state takes one graph scan. $\square$

If a chain cover with chain lengths $n_1, \dots, n_r$ is supplied, every ideal intersects each chain in a prefix. Consequently

$$J(P) \leq \prod_{i=1}^r (n_i + 1).$$

This gives a polynomial algorithm for each fixed number of covering chains, including the familiar product-state algorithm for a fixed list of strings. It is an XP-type bound in that parameter, not an assertion of fixed-parameter tractability with an exponent independent of r .

The verifier uses exhaustive legal-question minimax as a separate algorithm. For a nonempty possible-point mask A that is not monochromatic,

$$\begin{aligned} D(A) &= 1 + \min_U \max\{D(A \cap U), D(A \setminus U)\}, \\ L_{\text{tree}}(A) &= \min_U \{L_{\text{tree}}(A \cap U) + L_{\text{tree}}(A \setminus U)\}, \end{aligned}$$

where the minimum runs over open sets splitting A into two nonempty parts. A monochromatic nonempty mask has depth 0 and one leaf. The ideal automaton does not call these recurrences, making their agreement a useful independent finite check of the normal form.

24 Finite obstructions and bounded-budget decidability

Theorem 24.1 (Small finite witnesses of a failed layer budget). *For any poset P , finite or infinite, with q colors, if $L(P, c) > m$ for some $m \geq 1$, there is an induced finite colored subposet Q with*

$$|Q| \leq (m+1)q^m, \quad L(Q, c|_Q) > m. \quad (24.1)$$

Proof. For each word $v \in \Sigma^m$, Theorem 17.1 gives a reduced finite chain word not embedding into v . Stop its greedy embedding at the first failure. Each previous letter uses a strictly later position, so failure occurs within at most $m+1$ letters. Selecting the corresponding chain points gives an obstructing chain of at most that size.

Take the union of one such chain for each of the q^m words. If the induced subposet admitted a word of length at most m , append empty positions to obtain one of length exactly m . Its stored obstructing chain would then embed, a contradiction. This proves the bound. $\square$

In particular, a bounded finite query budget on an arbitrary Alexandrov poset is determined by its finite colored subposets. For a depth budget d , one may put $m = 2^d$ in (24.1). The numerical bound is an upper bound, not a claim of optimality or of a polynomial kernel.

On a finite DAG, enumerating the q^{2^d} words of length 2^d and applying Proposition 17.2 gives an exact algorithm with time

$$O\left(q^{2^d}(q2^d + n + e)\right).$$

Thus the problem is fixed-parameter tractable in the joint parameters q, d ; for fixed q , it is fixed-parameter tractable in the allowed query depth. This does not conflict with NP-completeness when the depth is part of an unbounded input.

Proposition 24.2 (Finite witnesses for sum incompatibility). *If every finite subfamily of a family of colored spaces has sum layer count at most m , then the entire sum has layer count at most m . If the bound fails, it fails already for a subfamily of at most q^m components.*

Proof. There are only q^m candidate words of length exactly m . If none is admissible on the whole sum, choose for each candidate a component on which it fails, using Proposition 21.1. Those at most q^m components already exclude every candidate. Padding an admissible shorter word shows that they also exclude every word of length at most m . $\square$

The two finite-witness statements have different domains. The induced-subposet statement is not a general finite-subspace theorem for Hausdorff spaces. Every finite subspace of a Hausdorff space is discrete, so it can have at most q necessary color layers, while the compact towers above have arbitrarily large layer complexity. Their obstructions lie in accumulation, not in finite Hausdorff subspaces.

25 Reproducible verification and certificates

The archive includes a standard-library Python verifier, not a proof-assistant formalization. Its full recorded run is, with the shipped name and an explicit output directory,

```
python3 code/02-finite-output-verify.py --max-n 5 --brute-n 4 --output <scratch>
```

A naturally labeled poset has every strict comparison directed from a smaller integer label to a larger one. Every finite poset admits such a labeling. The enumeration therefore covers every isomorphism type through the stated size, possibly many times; it is not an enumeration of unlabelled types.

For every binary and ternary coloring of every enumerated poset, the verifier compares exhaustive open-question minimax with shortest-word breadth-first search on ideals. It checks both minimum leaf count and the logarithmic depth formula, the binary formula when applicable, greedy feasibility, and the cyclic approximation bounds. The counts are:

Points	Naturally labeled posets	Binary colorings	Ternary colorings
1	1	2	3
2	2	8	18
3	7	56	189
4	40	640	3,240
5	357	11,424	86,751
Total	407	12,130	90,201

Thus there are **102,331 colored-poset tests**. Through four points, the program additionally enumerates all reduced chain words and solves their SCS problem by brute force. It checks **4,156 colored instances** in this way. For every candidate master word of lengths 1 through the point count in these instances, it compares chain subsequence containment, the greedy monotone map, and closed-set elimination. This gives **416,432 fixed-word checks**.

The restricted-start universal families are checked for $2 \leq q \leq 4$, $1 \leq m \leq 4$, and every $1 \leq s \leq q$: 36 parameter triples. For the 15 triples with $q \leq 3, m \leq 3$, the optimum is also found independently by exhaustive word enumeration. A further **1,176 tests** check exact common-prefix cancellation and the power-of-two reduction on pairs of small reduced ternary words.

The twelve-point example has a machine-readable certificate recording its predecessor masks, colors, master word, rank map, and balanced query tree. Every query is checked to be an upper set, and every point is followed to a correct leaf. The full outcomes are in `data/02-finite-output-verification.json`; tables and the strategy certificate are provided separately as CSV and JSON.

[Added 29 September 2026, batch 52] **Shipped names**. In the delivery the verifier was `code/verify.py`, its outputs lay in `results/`, and the command above read

```
python3 code/verify.py --max-n 5 --brute-n 4
```

Here `code/verify.py` is Part I's verifier, so the command above uses the shipped name. The recorded outputs are the five files and the console log

- `data/02-finite-output-verification.json`,
- `data/02-finite-output-strategy_certificate.json`,
- `data/02-finite-output-poset_counts.csv`,
- `data/02-finite-output-sharp_families.csv`,
- `data/02-finite-output-restricted_start_families.csv`,
- `data/02-finite-output-verification.log`.

The verifier writes unprefixed names; without `--output` it creates `results/` in the report directory. Its JSON files are written with the platform's line endings, so a rerun on Windows differs from the record by CRLF (and by the Python version and the elapsed time); its CSV files are CRLF on every platform, as delivered. The counts above agree with the recorded JSON and log.

These checks do not establish the infinite-space theorems, the general NP-hardness reduction, or historical novelty. Those claims rest respectively on the written proofs, the cited hardness input plus the written reduction, and a still-incomplete priority assessment. Finite Hausdorff subspaces are not used as substitutes for the compact tower arguments.

26 Proof dependencies, scope, and formalization

26.1 Proof dependency audit

The proofs form a short dependency chain. Tree flattening and balanced suffix search prove the normal form. The poset representation follows from monotone maps and finite greedy embeddings; compact realization follows from cofinite component neighborhoods. The adversarial next-occurrence argument proves the universal-word constant. These feed the synchronization theorem and the exact tower formula. Closed-set elimination supplies an independent topological normal-form test.

The only imported mathematical result needed for the NP-completeness claim is Theorem 22.2. The reduction to depth is proved here, including all boundary and input-size conditions. None of the topological or sharp-bound theorems depends on that complexity-theoretic input. No unproved conjecture is used as a theorem. The assertion that zero additive error cannot be guaranteed in polynomial time for three colors is, as usual, conditional on $P \neq NP$.

Indeed, Theorem 23.1 gives one extra query for three colors. A uniform guarantee of fewer than one extra query would, by integrality, be an exact algorithm and would contradict Theorem 22.4 if $P \neq NP$. In this precise integer-additive sense, the three-color approximation guarantee is optimal under that assumption. No analogous optimality claim is made for the bound $\lceil \log_2(q-1) \rceil$ when $q \geq 4$.

26.2 Relationship to the cited work

The binary repository layer framework is the direct starting point. Its binary counterexamples, binary alternating-chain algorithm, and binary Cantor–Bendixson results are not claimed anew. The finite-output exact tower formula, sharp synchronization family, and rounding-safe complexity transfer are the extensions developed here.

The SCS connection also has precursors in typed-graph fusion, as documented in [5]; a mere change of vocabulary would not be a priority claim. Likewise, Selivanov’s work places multi-valued partitions within a much broader descriptive-set-theoretic setting [6]. The present article is specifically about finite worst-case open-query budgets, explicit word certificates, and sharp constants. An independent comparison with the full partition-hierarchy and typed-fusion literature remains appropriate before submission as a novelty-certified paper.

26.3 A concrete proof-assistant plan

The finite part has a small trusted mathematical interface. A useful Lean formalization would proceed in the following order.

1. Define finite words, subsequence embeddings, reduced words, and color-respecting monotone maps to finite chains. Verify the greedy next-position recurrence and its least-map property.

2. Formalize finite binary query trees over a topology and the no-before-yes leaf order. Prove the suffix-open induction, then construct balanced suffix trees with the integer bound $m \leq 2^d$ rather than analytic logarithms.
3. Prove the chain-language equivalence, the universal-word upper bound, and the adversarial lower-bound witness. These yield synchronization with purely finite arithmetic at the final step.
4. Add a certified checker for a supplied word and rank map. A search program can remain untrusted: its positive certificates require only color agreement and monotonicity along input edges. Negative certificates can use an explicit failed chain for each candidate word or a verified ideal-state search, depending on the budget.
5. Formalize the convergent towers and their neighborhood property, then their rank-word profiles, the finite-height interval construction, and the exact tower invariant. The topological definitions should be kept separate from executable finite certificates.

The present archive contains no Lean file and makes no claim that any of these steps has been kernel-checked. The Python assertions are reproducible evidence about finite instances, not replacements for that formalization.

[Added 29 September 2026, batch 52] **Status in ProveIt.** None of these steps has been carried out in ProveIt, for Part II or for Part I: no Lean or Rocq file of the repository mentions open-query games, and the Lean library in the enclosing `SetTheory/Cardinals` directory is about large cardinals.

27 Further research questions and proposed directions

The following questions define a continuation program. They are not assertions that every formulation is a previously published open problem; their literature status should be checked individually before a priority claim.

Research question 1 (Exact invariants beyond convergent towers). For a general, possibly nonmetrizable or noncompact Hausdorff space of finite Cantor–Bendixson height, what local topological conditions make the uniform bound $(q - 1)h + s$ attainable? Can the invariant be characterized by the word profiles that accumulate at its nonisolated points?

The upper-bound proof uses only rank neighborhoods. Sharpness requires enough incompatible color patterns, and the tower construction supplies them by repeating each type cofinally. Separating those two ingredients is a concrete way to extend the classification without assuming that rank alone is always sufficient.

[Added 29 September 2026, batch 52] **Answered at $q = 2$; open for $q \geq 3$.** For two labels Part I answers both parts of this question. On every Hausdorff space of last derivative rank $h < \omega$ the bound $h + s$ is attained for $s = 1$ (the parity target of lemma 6.5) and, when $|X^{(h)}| \geq 2$, for $s = 2$ (the lower-bound proof of theorem 6.2); no condition beyond the Hausdorff property is needed, and $\text{sm}_2(X) = \text{sm}(X)$ depends only on h and on whether $X^{(h)}$ is a singleton, by (6.1). The question stands for $q \geq 3$.

Research question 2 (Small sharp synchronization families). What is the smallest number of reduced words of length at most m whose common supersequence length is $(q - 1)m + 1$? What is the smallest number needed merely to force the maximal depth $\lceil \log_2((q - 1)m + 1) \rceil$?

Our full family has $q(q - 1)^{m-1}$ words. For $m = 2$ the complete ordered-pair family is deletion-critical for its extremal layer count. For larger m , minimizing the obstruction family would substantially reduce both finite and compact examples.

Research question 3 (Optimal finite obstruction bounds). Determine the least function $B(q, m)$ such that every failure of an m -layer budget on a poset is witnessed by at most $B(q, m)$ points. How far can the bound $(m + 1)q^m$ be improved, and which colored posets are minimal obstructions?

The proof supplies explicit chains per candidate word but does not share their vertices efficiently. Critical SCS families are a natural first test class.

Research question 4 (Width and cover-graph parameters). For three colors, can the ideal-state algorithm be improved from an $(n + 1)^r$ -type bound for an r -chain cover to a fixed-parameter algorithm with parameter r ? Which cover-graph restrictions permit polynomial exact optimization when width is unbounded?

The finite-depth parameter is already fixed-parameter tractable by the word enumeration above. Width is different: long chains do not by themselves cause hardness, but synchronization among many chains does.

Research question 5 (One-extra-query algorithms with four or more colors). Does every fixed $q \geq 4$ admit a polynomial-time algorithm within one question of optimal, or is a larger additive gap necessary under an appropriate complexity assumption?

The cyclic construction gives $\lceil \log_2(q - 1) \rceil$ extra questions. Exact three-color hardness only excludes zero error; it does not prove the cyclic bound optimal for larger alphabets. Improved SCS approximation or a direct query-tree method could give a better result.

Research question 6 (Compact profile representations). Can the full word profile of a finite colored DAG be realized by a compact metrizable tower description of polynomial size without listing every chain word? Which sharing operations on descriptions preserve the profile exactly?

The current construction is exact but can duplicate exponentially many chains. A compositional representation of shared suborders would connect the topology more directly to symbolic verification and succinct optimization.

Research question 7 (Expected query cost and nonuniform costs). When points have a probability distribution, or queries have different costs, what replaces the layer-count logarithm? Can one optimize jointly over admissible layerings and ordered coding trees without enumerating all words?

Balancing by number of layers minimizes a worst-case bound, not expected cost. The flattening theorem still supplies an ordered structure, but the masses of the layers and the choice of word become essential data.

Research question 8 (Finite-output transfinite games). What analogue of the word profile describes well-founded, non-uniformly bounded query trees and games of ordinal length? Which synchronization laws survive at limit stages?

Neither $\lceil \log_2 L \rceil$ nor finite common-supersequence padding should be transferred mechanically to ordinal lengths. The treatment must specify the winning condition at limits and the kind of trees being ranked.

Research question 9 (Restricted query bases and noisy answers). How do the results change when only a designated basis of open sets may be queried, or when finitely many answers may be incorrect? What closure conditions on a query family retain a useful ordered-layer normal form?

The proof of Theorem 16.2 identifies the exact operations to investigate: finite intersections in a yes-branch and unions across a no-branch. Without these operations, the current equivalence need not persist.

Research question 10 (Certified integration into ProveIt). Can the finite greedy checker, the sharp-word adversary, and the compact rank-bound theorem be formalized as reusable interfaces rather than as isolated examples? Can proof-producing search return short negative certificates for practical three-color instances?

A successful integration would distinguish four outputs explicitly: a proved general theorem, a checked instance certificate, a computational experiment, and a literature-status claim. These are different forms of evidence.

28 Conclusion

The information needed to combine finite-output open-query algorithms is not just their depths. It is the set of color words in which their possible answers can be arranged with open suffixes. For posets this profile is an exact chain supersequence constraint; for compact convergent towers the same constraints are imposed by accumulation.

This representation yields three quantitative conclusions. First, the sharp uniform sum bound is $(q - 1)m + 1$ layers, with maximal depth increase $\lceil \log_2 q \rceil$. Second, finite Cantor–Bendixson height gives the sharp bound $(q - 1)h + s$, and the worst-case invariant of a finite sum of convergent towers is computed exactly by (20.2). Third, the introduction of a third label changes exact finite optimization from polynomial time to NP-completeness while still permitting a one-extra-query algorithm. All three conclusions have explicit witnesses and constructive proofs, and the finite part admits small, independently checkable certificates.

A Why countable compact Hausdorff spaces are scattered

This appendix supplies the fact used after corollary 10.3 without requiring the full ordinal classification of countable compact spaces.

Lemma A.1. *A nonempty compact Hausdorff space with no isolated points has at least $2^{\aleph_0}$ points.*

Proof. Construct nonempty open sets U_s indexed by finite binary strings. Start with the whole space. Inside each U_s , select two distinct points; this is possible because no nonempty open set is a singleton. By regularity, choose nonempty open neighborhoods U_{s0}, U_{s1} with disjoint closures contained in U_s .

For each infinite binary string z , the compact sets $\overline{U_{z \upharpoonright n}}$ are nonempty and nested, so their intersection is nonempty. Choose one point from each intersection. Distinct binary strings split at some level and hence give points in disjoint closed sets. This produces an injection of $2^{\mathbb{N}}$ into the space. $\square$

Proposition A.2. *Every countable compact Hausdorff space is scattered.*

Proof. If a space is not scattered, some nonempty subspace B has no isolated points. Its closure F has no isolated points either: an isolated point of F would, by density, belong to B and be isolated

in B . In a compact Hausdorff space, F is compact Hausdorff. The preceding lemma would make F uncountable, a contradiction. $\square$

This proof also shows why no additional set-theoretic hypothesis such as the continuum hypothesis is needed. Uncountability of $2^{\mathbb{N}}$ is sufficient.

B A proof audit and implementation checklist

The following points isolate possible sources of off-by-one or topology errors.

Zero questions. A constant target has $L = 1$, not $L = 0$, on a nonempty space. Thus its depth is $\lceil \log_2 1 \rceil = 0$. The empty space is assigned $\text{sm} = 0$ separately.

Derivative rank. The integer h is the index of the *last nonempty* derivative, not the height. The height is $h + 1$. A convergent sequence has $h = 1$ and height 2.

First closure. The general estimate is $F_n \subseteq X^{(n-1)}$ for $n \geq 1$, not $F_n \subseteq X^{(n)}$. The extra first step is exactly what creates the two oriented lengths $h + 1$ and $h + 2$ for a parity target.

Hausdorff separation. The lower bound with two terminal points uses disjoint open neighborhoods. The finite-height theorem is therefore stated for Hausdorff spaces. Its proof is not silently applied to arbitrary T_0 posets.

Arbitrary answers. The binary-search proof wins on every answer sequence, including one whose state becomes empty. In the ω -length certification strategy, an all-no certification path has empty final intersection. There is no hidden assumption that the Hider fixes a point beforehand.

Infinite sums. In the oriented finite-profile formula, an unbounded supremum of integers is recorded as ∞ , meaning “no finite decomposition”. The separate certification argument is needed to obtain the actual ordinal value ω .

Independent checking. The minimax implementation enumerates all open queries. It does not import the closed-layer answer as its return value. The mathematical proof and the finite computational check are deliberately distinct.

C Certificate and reproduction conventions (Part II)

In `data/02-finite-output-strategy_certificate.json` (Part I’s eight-point certificate is `data/strategy_certificate.json`), bit i of a mask denotes vertex i , with numbering starting at zero. The predecessor mask of a vertex contains *strict* predecessors. The supplied coloring consists of the chains 01, 02, 10, 12, 20, 21 in that order. The 1-based rank vector is

$$(1, 2, 1, 3, 2, 4, 2, 3, 3, 4, 3, 5).$$

At an internal tree node, query the upper set encoded by `open_mask`; follow `yes` or `no`. A leaf contains the reported color and its layer number. The word is 01201, with five layers.

The verifier requires Python 3.10 or later and no third-party packages. Do not use Python's `-O` option, which removes assertions. The default run is the recorded through-five-point run with through-four-point fixed-word checks. A smaller independent smoke test is

```
python3 code/02-finite-output-verify.py --max-n 3 --brute-n 3 --output <scratch>
```

The six-point option is substantially more expensive because both the poset count and the number of colorings grow quickly. Its availability is not a claim that a six-point run was performed for this article.

Build the PDF with two runs of

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The bibliography is embedded below; no BibTeX run, remote assets, or external figures are required. The archive also includes a Makefile and a source-status record. Font files are not distributed.

[Added 29 September 2026, batch 52] Shipped names. The smoke-test command above replaces the delivered `python3 code/verify.py --max-n 3 --brute-n 3 --output /tmp/open-query-smoke: code/verify.py` is Part I's verifier here, and `<scratch>` is any directory outside the report. The build paragraph describes the delivered standalone article. This merged report needs three `pdflatex` runs, as in section 12. The delivered Makefile and source-status record are shipped as `code/02-finite-output-Makefile` and `02-finite-output-RESEARCH_STATUS.md`, with the source audit `02-finite-output-SOURCES.md`. That Makefile is written for the delivery's layout; run from this directory it would call Part I's verifier and build this article in place, so it should not be used here.

References

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